B. Forward and Reverse Bias

Figure 3.26 (a) shows the energy band diagram of a *pn* junction in open circuit based on the above description (Figure 3.25). We ignored the vacuum level as this is not needed in the following discussion. When the *pn* junction is forward biased, the majority of the applied voltage drops across the depletion region so that the applied voltage is in opposition to the built-in potential, V_o. Figure 3.26 (b) shows the effect of forward bias which is to reduce the PE barrier from eV_o to $e(V_o - V)$. The electrons at E_c in the *n*-side can now readily overcome the PE barrier and diffuse to the *p*-side. The diffusing electrons from the *n*-side can be easily replenished by the negative terminal of the battery connected to this side. Similarly holes can now diffuse from the *p*- to *n*-side.

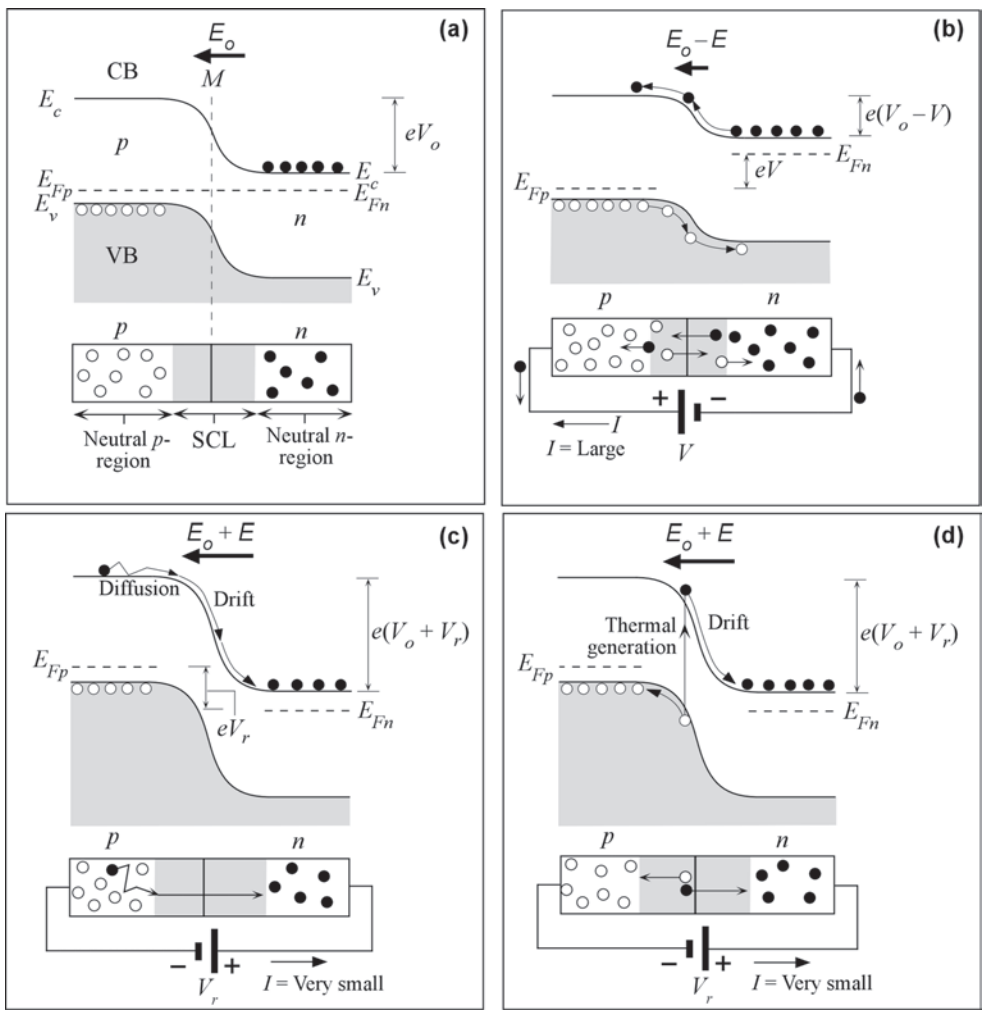


FIGURE 3.26 Energy band diagrams for a *pn* junction under (a) open circuit, (b) forward bias, and (c) and (d) reverse bias conditions. The reverse current in (c) is very small, usually negligible, since it depends on the diffusion of minority carriers to the depletion region. (d) Thermal generation of electron–hole pairs in the depletion region results in a small reverse current that is usually greater than the reverse current in (c).

The positive terminal of the battery can replenish those holes diffusing from the *p*- to the *n*-side. Therefore, a current flows through the junction and around the circuit.

The probability that an electron at E_c in the *n*-side overcomes the new PE barrier and diffuses to E_c in the *p*-side is now proportional to the Boltzmann factor $\exp\left[-e(V_o - V)/k_BT\right]$. The latter increases enormously even for small forward voltages. Thus, there is a net diffusion of electrons from the *n*- to *p*-side. Similar ideas also apply to holes at E_v in the *p*-side which also overcome the barrier $e(V_o - V)$ to diffuse into the *n*-side. Since the forward current is due to the number of electrons and holes overcoming the barrier, it is also proportional to $\exp\left[-e(V_o - V)/k_BT\right]$ or $\exp(eV/k_BT)$.

When a reverse bias, $V = -V_r$, is applied to the *pn* junction, the voltage again drops across the SCL. In this case, however, V_r adds to the built-in potential V_o so that the PE barrier becomes $e(V_o + V_r)$, as shown in Figure 3.26 (c). The field in the SCL at *M* increases to $E_o + E$ where E is the applied field (it is not simply V/W). The Shockley model predicts a small reverse saturation current due to the diffusion of minority carriers in neutral regions to the depletion region. Once they reach the depletion region, they will be drifted across by the large field and then collected by the battery. In the energy band diagram in Figure 3.26 (c), an electron in the *p*-side, within a diffusion length (L_e) to the depletion region, can diffuse to the SCL and then *fall down* the PE hill along E_c, over to the *n*-side where it will be collected by the battery. The process of falling down a PE hill is the same process as being driven by a field, in this case by $E_o + E$. The electron lost from the *p*-side is replenished from the negative terminal of the battery (or by thermal generation) to maintain the small reverse current. The same idea applies to the hole in the *n*-side within a diffusion length (L_h) to the SCL. Such minority carrier diffusion is the essence of the Shockley model. The reverse current magnitude is I_{so}. The measured reverse current in Ge diodes at room temperature follow the Shockley model as can be seen in Figure 3.22 (b).

In the case of many semiconductors, however, there is a more significant contribution to the reverse current arising from the thermal generation of electron–hole pairs in the SCL, as shown in Figure 3.26 (d), where the field here separates the pair. The electron falls down the PE hill, down E_c, to the *n*-side to be collected by the battery. Similarly the hole falls down its own PE hill (energy increases downwards for holes) to make it to the *p*-side.

EXAMPLE 3.9.1 The built-in voltage from the band diagram

Derive the expression for the built-in voltage V_o using the energy band diagram in Figure 3.25.

Solution

The extent of band bending, going from (a) to (b) in Figure 3.25, gives the PE barrier eV_o, thus

$$eV_o = \Phi_p - \Phi_n = E_{Fn} - E_{Fp} \text{ (before contact)}$$

Before the contact, on the *n*-side we have

$$n = N_c \exp\left[-(E_c - E_{Fn})/k_BT\right] = N_d$$

so that

$$E_c - E_{Fn} = -k_BT \ln(N_d/N_c) \tag{3.9.1}$$

On the *p*-side

$$n = N_c \exp\left[-(E_c - E_{Fp})/k_BT\right] = n_i^2/N_a$$

so that

$$E_c - E_{Fp} = -k_B T \ln\left[n_i^2/(N_a N_c)\right] \tag{3.9.2}$$

Thus, subtracting Eq. (3.9.2) from (3.9.1) gives

Built-in potential V_o

$$eV_o = E_{Fn} - E_{Fp} = k_B T \ln\left[(N_a N_d)/n_i^2\right] \tag{3.9.3}$$

3.10 HETEROJUNCTIONS

The *pn* junction with the band diagram in Figure 3.25 is a junction within the same crystal (Si), and hence the bandgap does not change along the device; it represents a **homojunction**. A **heterojunction** is a junction between two different semiconductor crystals with different bandgaps E_{g1} and E_{g2}, for example, between GaAs and Al_xGa_{1-x} As ternary alloys. Inasmuch as the bandgaps are now different, their alignment becomes important. If the bandgap difference is $\Delta E_g = E_{g2} - E_{g1}$, then this difference is taken up by a difference $\Delta E_c (= E_{c2} - E_{c1})$ in the CB edges, and $\Delta E_v (= E_{v1} - E_{v2})$ in the VB edges. The energy discontinuities in ΔE_c and ΔE_v are called **band offsets** and play an important role in heterojunction devices.

The terms *heterojunction* and *heterostructure* are frequently used interchangeably, though a heterostructure usually has more than one heterojunction. There may or may not be a change in the doping across the heterojucntion. The doping in the wider bandgap semiconductor is usually denoted with a capital letter *N* or *P*, and that in the narrower bandgap semiconductor with lower case *n* or *p*. There are two cases of particular importance, called Type I and Type II heterojunctions. In a Type I **straddled bandgap alignment heterojunction**, as illustrated in Figure 3.27 (a), the smaller bandgap material offers the lowest energy for both electrons and holes as in GaAs/Al_xGa_{1-x}As heterostructures in which GaAs has a smaller bandgap than Al_xGa_{1-x}As. Tpe I is the most common heterostructure in optoelectronic devices, for example, Ga_xIn_{1-x}As/InP, GaAs/Ga_xIn_{1-x}P. In a Type II **staggered lineup heterojunction**, as illustrated

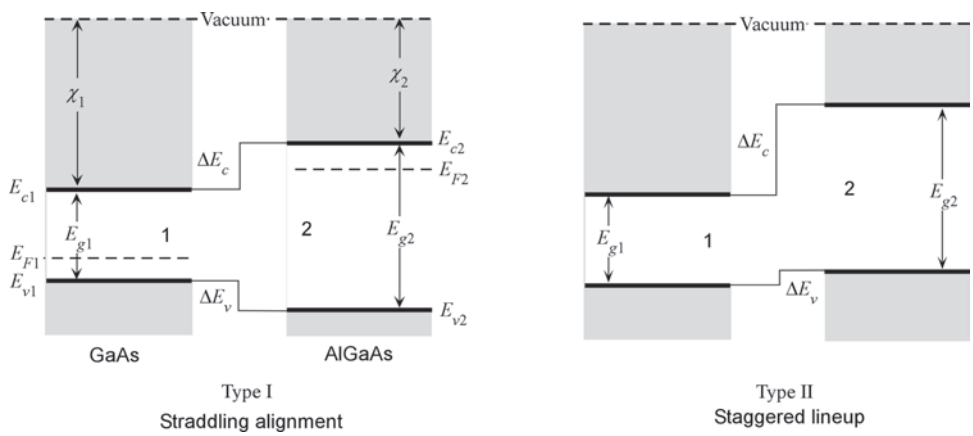


FIGURE 3.27 Two types of heterojunction and the definitions of band offsets Type I and Type II between two semiconductor crystals 1 and 2. Crystal 1 has a narrower bandgap E_{g1} than E_{g2} for crystal 2. Note that the semiconductors are not in contact so that the Fermi level in each is different. In the Type I example, crystal 1 (GaAs) is *p*-type and crystal 2 (AlGaAs) is *N*-type. (Note that the subscripts 1 and 2 refer to semiconductors on the left and right respectively.)

in Figure 3.27 (b), minimum energies for holes and electrons are in the different materials. $Ga_xIn_{1-x}As/GaAs_ySb_{1-y}$ heterojunctions over wide compositions follow the Type II behavior.

The band edge profiles for a given heterostructure are determined by the doping levels, carrier transport, and recombination around the junction, as they are for a simple *pn*-homojunction. The open-circuit heterojunction band diagrams are straightforward based on the principle that the Fermi level must be uniform once a contact is made, and we know how ΔE_g is shared between ΔE_c and ΔE_v. Figure 3.28 (a) shows the energy band diagram of an *Np* heterojunction between *n*-type AlGaAs and *p*-type GaAs. First, notice that E_F is uniform through the device as an equilibrium requirement. Far away from the junction on the *N*-side, we have an *n*-type wide bandgap AlGaAs with E_F close to E_c. Far away from the junction on the right, we have a *p*-type narrower bandgap GaAs with E_F close to E_v. In the depletion regions, around the junction, E_c and E_v must bend because there is an internal field, as we know from the ordinary *pn* junction. E_{c1} and E_{v1} of AlGaAs bend upwards and E_{c2} and E_{v2} of GaAs bend downwards as in the normal *pn* junction. We need to join E_{v1} to E_{v2} but also need to account for ΔE_v, which is easily done by putting ΔE_v between E_{v1} and E_{v2} at the junction as shown in Figure 3.28 (a). Similarly, we need to join E_{c1} to E_{c2} but also need to account for ΔE_c. In this case, we can only join E_{c1} and E_{c2} by having a narrow "spike" whose height must be ΔE_c at the junction as shown in Figure 3.28 (a).

One obvious conclusion is that the potential barrier for hole injection, $(E_{v2} - E_{v1})$, from *p* to *N* is greater than the barrier $(E_{c2} - E_{c1})$ for electron injection from *N* to *p*. Under forward bias, the current will be dominated by the injection of electrons from the *N*-side into the *p*-side. ΔE_v has increased the potential barrier against holes and ΔE_c has decreased the potential barrier against electrons.

The energy band diagram for a *pP*-type heterojunction is shown in Figure 3.28 (b). The basic principle for drawing the diagram is the same as before. In this case, there is a small spike

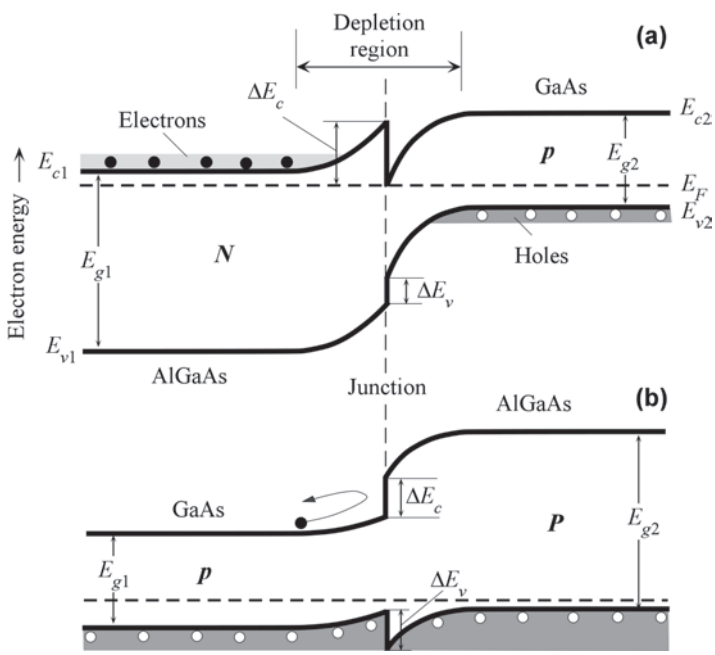


FIGURE 3.28 (a) *nP* and (b) *pP* heterojunctions and their energy band diagrams (schematic only to illustrate general features). Under open circuit and equilibrium conditions, the Fermi level E_F must be uniform, that is, continuous throughout the device. If E_F is close to the conduction band edge, E_c, it results in an *n*-type, and if it is close to the valence band edge, E_v, it results in a *p*-type semiconductor. There is a discontinuity ΔE_c as in E_c, and ΔE_v in E_v, right at the junction.

in E_v at the junction. ΔE_c increases the potential barrier from E_{c1} to E_{c2}. An electron in the CB in the *p*-side cannot simply overcome this barrier and enter the *P*-side. Holes in the VB of *p*- and *P*-sides can easily cross through the spike by tunneling. Both the *Np* and *pP* heterojunctions are used extensively in LED and semiconductor laser diode heterostructures.

3.11 LIGHT-EMITTING DIODES: PRINCIPLES

A. Homojunction LEDs

A light-emitting diode is essentially a *pn* junction diode typically made from a direct bandgap semiconductor, for example, GaAs, in which the electron–hole pair recombination results in the emission of a photon. The emitted photon energy is therefore approximately equal to the bandgap energy, $h\upsilon \approx E_g$. Figure 3.29 (a) shows the energy band diagram of an unbiased pn^+ junction device in which the *n*-side is more heavily doped than the *p*-side. The band diagram is drawn to keep the Fermi level, E_{Fp} and E_{Fn} on the *p*- and *n*-sides, uniform through the device which is a requirement of equilibrium with no applied bias. The depletion region in a pn^+ device extends mainly into the *p*-side. There is a potential energy (PE) barrier eV_o from E_c on the *n*-side to E_c on the *p*-side, that is, $\Delta E_c = eV_o$, where V_o is the *built-in voltage.* The higher concentration of conduction (free) electrons in the *n*-side encourages the diffusion of these electrons from the *n*- to the *p*-side. This net electron diffusion, however, is prevented by the electron PE barrier eV_o.

As soon as a forward bias *V* is applied, this voltage drops almost entirely across the depletion region since this is the most resistive part of the device. Consequently, the built-in potential V_o is reduced to $V_o - V$, which then allows the electrons from the n^+ side to diffuse, or become injected, into the *p*-side as illustrated in Figure 3.29 (b). The hole injection component from *p* into the n^+ side is much smaller than the electron injection component from the n^+ to *p*-side. The recombination of injected electrons in the depletion region as well as in the neutral *p*-side results in the *spontaneous emission* of photons. Recombination primarily occurs within the depletion region and within a volume extending over the diffusion length L_e of the electrons in the *p*-side. (Electron injection is preferred over hole injection in GaAs LEDs because electrons have a higher mobility and hence a larger diffusion coefficient.) This recombination zone is

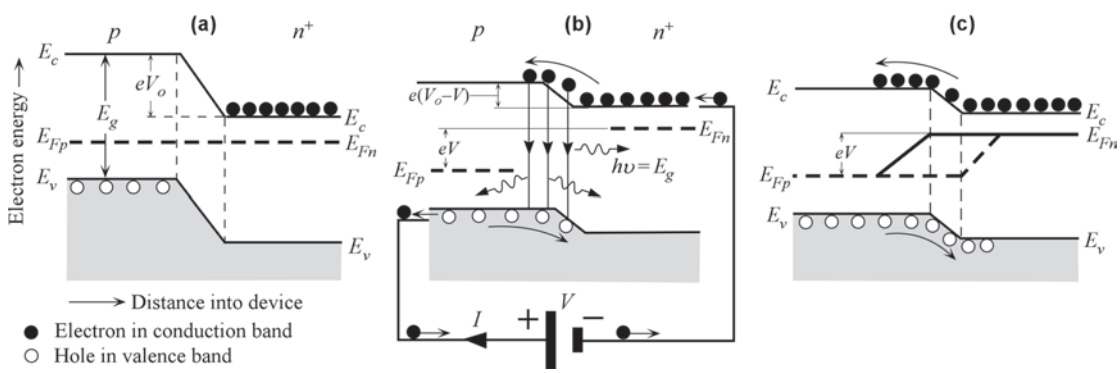


FIGURE 3.29 (a) The energy band diagram of a pn^+ (heavily *n*-type doped) junction without any bias. Built-in potential V_o prevents electrons from diffusing from n^+ to *p*-side. (b) The applied bias potential *V* reduces V_o and thereby allows electrons to diffuse, be injected, into the *p*-side. Recombination around the junction and within the diffusion length of the electrons in the *p*-side leads to spontaneous photon emission. (c) Quasi-Fermi levels E_{Fp} and E_{Fn} for holes and electrons across a forward-biased *pn*-junction.

frequently called the **active region**.[16] The phenomenon of light emission from EHP recombination as a result of minority carrier injection, as in this case, is called **injection electroluminescence**.[17] Because of the statistical nature of the natural recombination process between electrons and holes, the emitted photons are in random directions; such randomly emitted radiation is called **spontaneous emission**. The LED structure has to be such that the emitted photons can escape the device without being reabsorbed by the semiconductor material. This means the *p*-side has to be sufficiently narrow, or we have to use heterostructure devices as discussed below. Further, the rate of recombination depends on the product *np*, which must be increased to enhance the emitted photon flux. We need to confine the carriers and increase the carrier concentration, again achievable by using heterostructures.

Notice that in Figure 3.29 (b), under forward bias, E_{Fn} and E_{Fp} are separated by eV inasmuch as the electrical work done per electron, eV, must be ΔE_F or $E_{Fn} - E_{Fp}$. Also notice that E_{Fn} and E_{Fp} extend into the depletion region, where there are now two Fermi levels as highlighted in Figure 3.29 (c). We can use E_{Fn} to represent the electron concentration through the general expression $n = N_c \exp\left[-(E_c - E_{Fn})/k_B T\right]$, not only on the *n*-side but also in the depletion region, and even in the *p*-side; E_{Fn} is then called a **quasi-Fermi level for electrons**. Similarly, we can use E_{Fp} as a **quasi-Fermi level for holes** and use $p = N_v \exp\left[-(E_{Fp} - E_v)/k_B T\right]$ to represent the hole concentration.[18] In the depletion region, E_{Fn} and E_{Fp} are separated within the same spatial region so that $np > n_i^2$, which is expected given that we are injecting minority carriers and have non-equilibrium. Figure 3.29 (c) shows the variation of E_{Fp} and E_{Fn} across a biased *pn* junction. Such variation in E_{Fp} and E_{Fn} will result in $p(x)$ and $n(x)$ profiles across the device that match the sketch in Figure 3.16 (a) for a pn^+ junction; we have effectively represented carrier concentration profiles in terms of E_{Fp} and E_{Fn} on the energy band diagram. Notice that E_{Fn} on the *p*-side slopes down, away from E_c, to represent the decrease in the electron concentration in the neutral *p*-side, due to recombination of injected electrons with the majority carriers (holes). E_{Fn} reaches E_{Fp} over a distance that is very roughly the diffusion length of injected electrons. The region where E_{Fn} and E_{Fp} have merged and are the same represents the neutral bulk *p*-side. Similar arguments apply to holes injected into the *n*-side. E_{Fp} slopes and merges with E_{Fn} over a distance that is roughly the hole diffusion length (shorter than the electron diffusion length).

It is clear that the emission of photons in the LED involves the direct recombination of electrons and holes; that is, an electron makes a radiative transition from the bottom of the CB to the top of the VB, where there is an empty state (a hole) as in Figure 3.14 (a). It may therefore be thought that one should avoid using indirect semiconductors in LEDs. However, this is not entirely true because it is possible to introduce impurities into certain indirect semiconductors and thereby enable radiative transitions through these impurities. A good example is GaP doped with N, that is, GaP:N, which are used in inexpensive green LEDs in numerous indicator and display applications in consumer optoelectronics. GaP is an indirect bandgap semiconductor with E_g of 2.26 eV (corresponding to green light at 550 nm). When N is added, N substitutes for P,

[16] The "active region" term is probably more appropriate for laser diodes in which there is photon amplification in this region. However, the term is also used in LED discussions.

[17] In 1907, Henry Round, working for Guglielmo Marconi in England, observed yellow light emission when a current was passed through a metal-SiC (silicon carbide or corborandum) crystal rectifier. Oleg Losev in the former USSR carried out a number of experiments on the emission of light upon passing a current through such metal-SiC rectifiers during the mid-1970s. He reported his systematic experiments in a number of journals, called the effect the "inverse photoelectric effect," and proposed that emission frequency is $\upsilon = eV/h$. (See Nikolay Zheludev, *Nat. Photonics*, *1*, 189, 2007.)

[18] Clearly, quasi-Fermi levels provide a convenient and useful means of representing carrier concentrations on the energy diagram.

and is called an **isoelectronic dopant**. However, N dopants create recombination centers that have a localized energy level at E_r, roughly $0.1 - 0.2$ eV below the CB as shown in Figure 3.14 (c). An electron at the bottom of the CB can easily fall into E_r during which phonons are emitted to make up for the change in the momentum from k_{CB} to k_{VB}. The electron at E_r can then drop down to the top of the VB and emit a photon $h\upsilon = E_r - E_v$, with $h\upsilon$ slightly below E_g. Since $h\upsilon$ is less than E_g, there is less likelihood of reabsorbing the photon in the bulk of GaP, especially if N doping is introduced in the junction (recombination) region only.

B. Heterostructure High Intensity LEDs

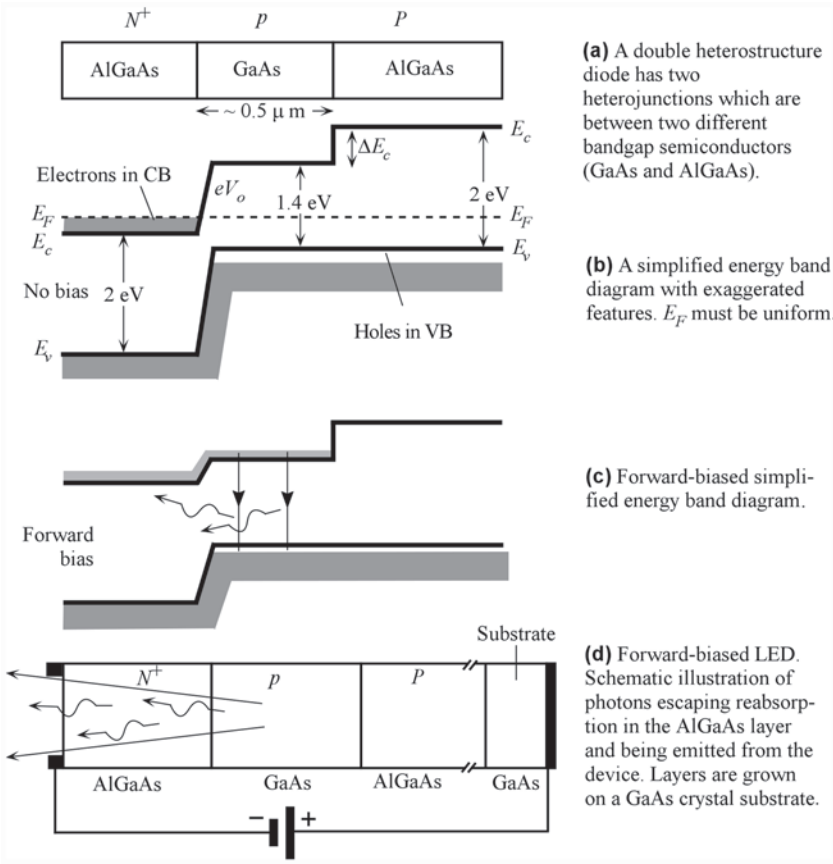
The *pn*-junction LED shown in Figure 3.29 suffers from a number of drawbacks and has a low efficiency. The *p*-region must be narrow to allow the photons to escape without much reabsorption. When the *p*-side is narrow, some of the injected electrons in the *p*-side reach the surface by diffusion and recombine through crystal defects near the surface. This radiationless recombination process decreases the light output. In addition, if the recombination occurs over a relatively large volume (or distance), due to long electron diffusion lengths, then the chances of reabsorption of emitted photons becomes higher; the amount of reabsorption increases with the material volume.

On the other hand, heterostructure LEDs can have exceptionally high efficiencies along with a number of other advantages; most modern LEDs are heterostructure devices (HDs). First, we note that the refractive index of a semiconductor material depends on its bandgap. A wider bandgap semiconductor has a lower refractive index. This means that by constructing LEDs from heterostructures, we can engineer a dielectric waveguide within the device and thereby channel photons out from the recombination region.

LED constructions for increasing the intensity of the output light make use of the double heterostructure (DH) structure. Figure 3.30 (a) shows a **double-heterostructure** (DH) device based on two heterojunctions between different semiconductor crystals with different bandgaps. In this case the semiconductors are AlGaAs ($Al_xGa_{1-x}As$) with $E_g \approx 2$ eV and GaAs with $E_g \approx 1.4$ eV. The double heterostructure in Figure 3.30 (a) has an N^+p heterojunction between N^+-AlGaAs and *p*-GaAs. There is another heterojunction between *p*-GaAs and *P*-AlGaAs. The *p*-GaAs region is a thin layer, typically a fraction of a micron and it is lightly doped. The reader would have noticed that this HD device is made of the heterojunctions in Figure 3.28 (a) and (b) one after the other to form an *N-p-P* device.

The simplified energy band diagram for the whole device in the absence of an applied voltage is shown in Figure 3.30 (b). The Fermi-level E_F is continuous through the whole structure. There is a potential energy barrier eV_o for electrons in the CB of N^+-AlGaAs against diffusion into *p*-GaAs. The spike in E_c at the junction, as in Figure 3.28 (a), is ignored in this simplified sketch. There is a bandgap change at the junction between *p*-GaAs and *P*-AlGaAs which results in a step change ΔE_c in E_c between the two CBs of *p*-GaAs and *P*-AlGaAs as in Figure 3.28 (b). This step change, ΔE_c, is effectively a potential energy barrier that prevents any electron in the CB in *p*-GaAs passing into the CB of *P*-AlGaAs.

When a forward bias is applied, the majority of this voltage drops between the N^+-AlGaAs and *p*-GaAs (*i.e.,* across the depletion region) and reduces the potential energy barrier eV_o, just as in the normal *pn* junction diode. This allows electrons in the CB of N^+-AlGaAs to be injected (by diffusion) into *p*-GaAs as shown in Figure 3.30 (c). These electrons, however, are *confined* to the CB of *p*-GaAs since there is a barrier ΔE_c between *p*-GaAs and *P*-AlGaAs (see Figure 3.28 (b)). The wide bandgap AlGaAs layer therefore acts as a **confining layer** that restricts injected electrons to the *p*-GaAs layer. The recombination of injected electrons with the holes present in



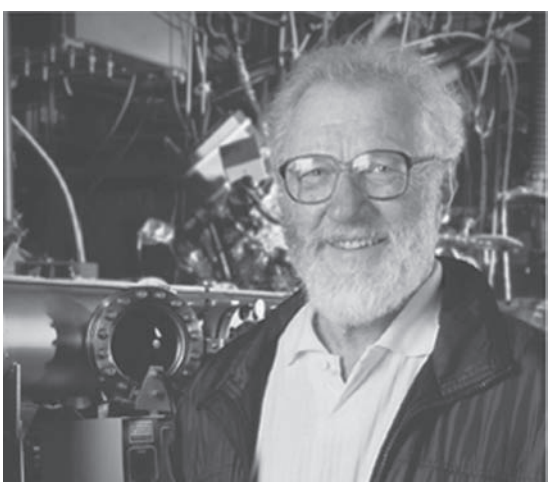
(a) A double heterostructure diode has two heterojunctions which are between two different bandgap semiconductors (GaAs and AlGaAs).

(b) A simplified energy band diagram with exaggerated features. E_F must be uniform.

(c) Forward-biased simplified energy band diagram.

(d) Forward-biased LED. Schematic illustration of photons escaping reabsorption in the AlGaAs layer and being emitted from the device. Layers are grown on a GaAs crystal substrate.

FIGURE 3.30 The double heterostructure AlGaAs/GaAs/AlGaAs LED structure (highly simplified schematic).



Herbert Kroemer (left), along with Zhores Alferov (Chapter 4 opening photo), played a key role in the development of semiconductor heterostructures that are widely used in modern optoelectronics.[19] Herbert Kroemer was also well-recognized for his experimental work on the fabrication of heterostructures by using an atomic layer-by-layer crystal growth technique called Molecular Beam Epitaxy (MBE)—the equipment shown behind Professor Kroemer in the photo. Since 1976, Professor Kroemer has been with the University of California, Santa Barbara, where he continues his research. Herbert Kroemer and Zhores Alferov shared the Nobel Prize in Physics (2000) with Jack Kilby. Their Nobel citation is "*for developing semiconductor heterostructures used in high-speed- and opto-electronics.*" *(Courtesy of Herbert Kroemer, University of California, Santa Barbara.)*

[19] See H. Kroemer, *Rev. Mod. Phys.*, *73*, 783, 2001.

this p-GaAs layer results in spontaneous photon emission. Since the bandgap E_g of AlGaAs is greater than that of GaAs, the emitted photons do not get reabsorbed as they escape the active region and can reach the surface of the device as illustrated in Figure 3.30 (d). Since light is also not absorbed in P-AlGaAs, it can be *reflected* to increase the light output, by, for example, using a dielectric mirror at the end of P-AlGaAs. Another advantage of the AlGaAs/GaAs heterojunction is that there is only a small lattice mismatch between the two crystal structures and hence negligible strain induced interfacial defects (*e.g.*, dislocations) in the device compared with the defects at the surface of the semiconductor in conventional homojunction LED structure. The DH LED is much more efficient than the homojunction LED.

C. Output Spectrum

The energy of an emitted photon from an LED is not simply equal to the bandgap energy E_g because electrons in the conduction band are distributed in energy and so are the holes in the valence band. Suppose that the active region is p-type, and excess electrons have been injected by forward bias. Figure 3.31 (a) and (b) illustrates the energy band diagram and the energy distributions of electrons and holes in the CB and VB, respectively, for a p-type semiconductor. The electron concentration as a function of energy in the CB is given by $g(E)f(E)$, where $g(E)$ is the density of states in the CB and $f(E)$ is the Fermi–Dirac function (probability of finding an electron in a state with energy E). The product $g(E)f(E)$ represents the electron concentration per unit energy, $n_E(E)$, and is plotted along the horizontal axis in Figure 3.31 (b). There is a similar energy distribution for holes, p_E, in the VB but p_E is enormously larger than n_E. The E–k diagram for a typical direct bandgap semiconductor (such as GaAs) is shown in Figure 3.31 (c). Since the hole concentration is very large, we can assume that the rate of recombination will depend primarily on the concentration of injected electrons.[20] The electron concentration in the CB

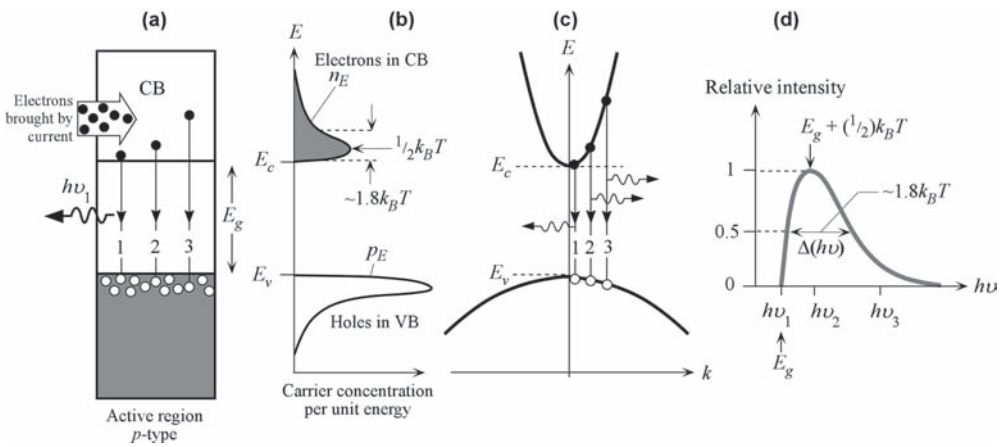


FIGURE 3.31 (a) Energy band diagram with possible recombination paths. (b) Energy distribution of electrons in the CB and holes in the VB. The highest electron concentration is $(1/2)k_BT$ above E_c. (c) A simplified E–k (equivalent to energy vs. momentum) diagram and direct recombination paths in which k (*i.e.,* momentum) is conserved. (d) The relative light intensity as a function of photon energy based on (b) and (c).

[20] We also need to consider the quantum mechanical transition probability from an occupied state in the CB to an empty state in the VB but, for simplicity, we will take this to be constant, that is, we will ignore it.

as a function of energy is asymmetrical, and has a peak at $\frac{1}{2}k_BT$ above E_c. The energy spread of these electrons is about $1.8k_BT$ between the half-maximum points, as in Figure 3.31 (b), or roughly $\sim 2k_BT$. When an electron at E_c recombines with a hole at E_v, shown as the transition 1 in Figure 3.31 (a), a photon is emitted with an energy $h\upsilon_1 = E_c - E_v = E_g$. Since there are not many electrons and holes at the band edges, this type of recombination does not occur frequently, and the emitted light intensity from a type 1 transition is small.

The transition that involves the largest electron concentration (the peak in n_E) is shown as 2 in Figure 3.31 (a), and emits a photon with $h\upsilon_2 > h\upsilon_1$. Such transitions occur frequently (large n_E), and hence the emitted intensity from type 2 transitions is much larger than that for type 1. Similarly, the transition marked 3, corresponding to $h\upsilon_3 > h\upsilon_2$, involves an electron quite high up in the CB where n_E is very small. Such type 3 transitions are infrequent and lead to a small emission intensity. The emission intensity therefore rises to a maximum and then falls with $h\upsilon$ as shown in Figure 3.31 (d).

One might conclude that the highest emitted intensity should intuitively correspond to the transition from the peak in n_E to the peak in p_E in Figure 3.31 (b) that emits $h\upsilon = E_g + k_BT$. However, we also need to consider the conservation of momentum, and hence the E–k diagram in Figure 3.31 (c). The momentum of the emitted photon is negligibly small, so that an electron falls straight down in the E–k diagram without changing its k-vector, that is, the electron momentum $\hbar k$ is conserved. The E–k curvatures are different in the CB and the VB. The electron at $E_c + \frac{1}{2}k_BT$ cannot just recombine with the hole at $E_v + \frac{1}{2}k_BT$ because that transition does not satisfy the $\hbar k$-conservation. As shown in Figure 3.31 (c), direct recombination involves energetic electrons spreading over several k_BT in the CB, more than the holes in the VB, because the E–k curvature is narrower in the CB and broader in the VB. It is apparent that the emission spectrum in this case is determined by n_E, the energy spread in the electrons in the CB, so that the emission has a peak at roughly $E_g + \frac{1}{2}k_BT$. Further the spread $\Delta(h\upsilon)$ in the emitted photon energies should be roughly the spread in n_E, that is, $\Delta(h\upsilon) \approx 1.8k_BT$.

The intuitive relative light intensity vs. photon energy characteristic of the output spectrum based on n_E and the E–k diagram is shown in Figure 3.31 (d) and represents an important LED characteristic. Given the spectrum in Figure 3.31 (d), we can also obtain the relative light intensity vs. wavelength characteristic since $\lambda = c/\upsilon$, which would look like Figure 3.31 (d) flipped horizontally. The **linewidth** of the output spectrum, $\Delta\upsilon$ or $\Delta\lambda$, is defined as the width between half-intensity points as illustrated in Figure 3.31 (d).

The experimentally observed output spectrum, that is, the relative intensity vs. wavelength characteristics, from an LED depends not only on the semiconductor material, including dopant concentrations, but also on the structure of the *pn* junction diode. The spectrum in Figure 3.31 (d) represents a highly idealized spectrum without including the effects of heavy doping on the energy bands. For a heavily doped *n*-type semiconductor there are so many donors that the electron wavefunctions at these donors overlap to generate a narrow impurity band centered at E_d but extending into the conduction band. Thus, the donor impurity band overlaps the conduction band and hence effectively lowers E_c as in Figure 3.10 (a). The minimum emitted photon energy from heavily doped semiconductors is therefore less than E_g and depends on the amount of doping.

Typical output spectrum from an AlGaAs IR LED is shown in Figure 3.32 (a). Notice that the spectrum exhibits significantly less asymmetry than the idealized spectrum in Figure 3.31 (d). The width of the spectrum is about 40 nm, which corresponds to a width of about $2.9k_BT$ in the energy distribution of the emitted photons, more than the expected $1.8k_BT$. The reasons for not observing the theoretical spectrum in Figure 3.31 (d) are essentially twofold. First, higher energy photons become reabsorbed in the material and photogenerate electrons and holes.

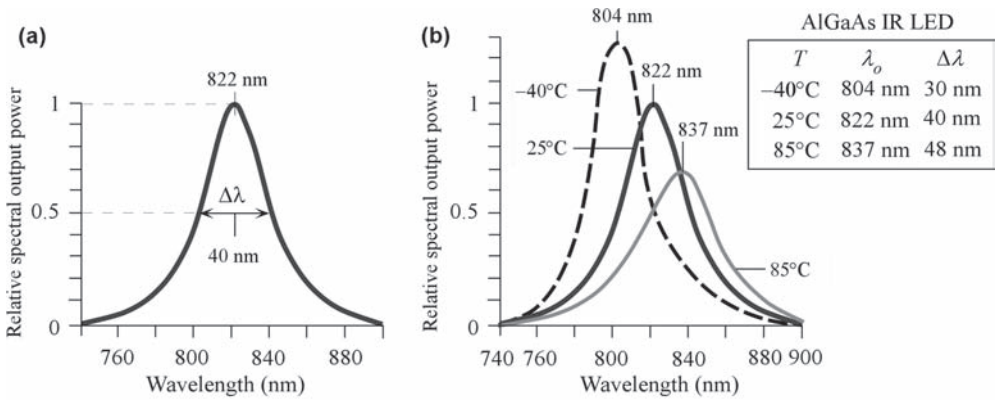


FIGURE 3.32 (a) A typical output spectrum (relative intensity vs. wavelength) from an IR (infrared) AlGaAs LED. (b) The output spectrum of the LED in (a) at three temperatures: 25°C, −40°C, and 85°C. Values normalized to peak emission at 25°C. The spectral widths are FWHM.

These electrons and holes thermalize and end up recombining to emit photons with lower energies, closer to E_g. Thus photons become redistributed. Secondly, the band edges E_c and E_v are not sharp in heavily doped semiconductors, which leads to the smearing of the well-defined E_g for the emission onset. The peak emission frequency υ_o and the spectral width $\Delta\upsilon$ in photon energy in LEDs with a direct bandgap active region are normally described by

LED spectrum in frequency

$$h\upsilon_o \approx E_g + \tfrac{1}{2}k_BT \quad \text{and} \quad h\Delta\upsilon = mk_BT \tag{3.11.1}$$

where m is a numerical factor that is typically between 1.5 and 3.5, though for many LEDs, $m \approx 3$ is a good value (see Question 3.23). The corresponding peak wavelength λ_o and the spectral width $\Delta\lambda$ can be easily found from Eq. (3.11.1) as in Example 3.11.1. Further, the actual position of the peak is likely to be somewhat more than $\frac{1}{2}k_BT$ in Eq. (3.11.1), given a broader observed spread than $1.8k_BT$ expected from the theory. As the temperature increases, the change in $h\upsilon_o$ in Eq. (3.11.1) is due mainly to the decrease in the bandgap E_g with temperature. The peak emission wavelength λ_o, corresponding to υ_o, therefore increases with temperature as shown Figure 3.32 (b). In addition, the linewidth $\Delta\lambda$ becomes longer at higher temperatures as electrons are distributed further into the CB. Thus, a wider spectrum of photon energies are emitted upon EHP recombination. The variation of the bandgap E_g with temperature is commonly represented by the **Varshni equation**

Varshni bandgap equation

$$E_g = E_{go} - \frac{AT^2}{B + T} \tag{3.11.2}$$

where E_{go} is the bandgap at $T = 0$ K, and A and B are material-specific constants that are listed in various semiconductor handbooks.

Equation (3.11.1) does not apply to an indirect bandgap semiconductor in which a recombination center is involved in the radiative transition, such as GaP:N. The electron localized at the recombination center would have a significant uncertainty in its momentum Δp and hence an uncertainty ΔE in its energy (Heisenberg's uncertainty principle, $\Delta p\Delta x \sim \hbar$). The emitted photon spectrum depends on this ΔE, and is wider than $3k_BT$ that is involved in the direct recombination process in Figure 3.31.

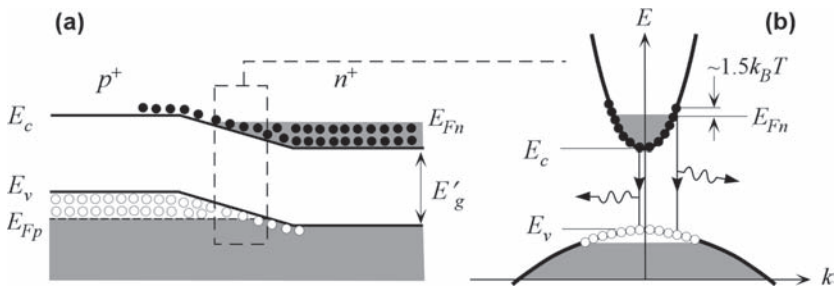


FIGURE 3.33 (a) Forward-biased degenerately doped *pn* junction. The bandgap E_g' is narrower than that in the undoped bulk crystal. The quasi-Fermi levels E_{Fn} and E_{Fp} overlap around the junction. (b) The transitions involved in a degenerately doped *pn* junction.

For degenerately doped junctions, the Fermi level E_{Fn} on the *n*-side and E_{Fp} on the *p*-side will be in the CB and VB, respectively. Under a large forward bias, the active region can have E_{Fn} in the CB, E_{Fp} in the VB around the junction as shown in Figure 3.33 (a). The bandgap E_g' is narrower than E_g in the undoped crystal (see Figure 3.10 (a) and (b)). As shown in the E–k diagram in Figure 3.33 (b), electrons occupy states from the CB edge E_c up to about $\sim 1.5k_BT$ above E_{Fn}. The emission spectrum will extend from $h\upsilon \approx E_g$ to about $E_g + E_{Fn} + 1.5k_BT$ so the width is roughly $E_{Fn} + 1.5k_BT$.

EXAMPLE 3.11.1 LED spectral linewidth

We know that a spread in the output wavelengths is related to a spread in the emitted photon energies as illustrated in Figure 3.31. The emitted photon energy $h\upsilon = hc/\lambda$. Assume that the spread in the photon energies $\Delta(h\upsilon) \approx 3kT$ between the half intensity points. Show that the corresponding linewidth $\Delta\lambda$ between the *half intensity points* in the output spectrum is

$$\Delta\lambda = \lambda_o^2 \frac{3kT}{hc} \tag{3.11.3}$$

LED spectral linewidth

where λ_o is the peak wavelength. What is the spectral linewidth of an optical communications LED operating at 1310 nm and at 300 K?

Solution

First consider the relationship between the photon frequency υ and λ,

$$\lambda = \frac{c}{\upsilon} = \frac{hc}{h\upsilon}$$

in which $h\upsilon$ is the photon energy. We can differentiate this

$$\frac{d\lambda}{d(h\upsilon)} = -\frac{hc}{(h\upsilon)^2} = -\frac{\lambda^2}{hc} \tag{3.11.4}$$

The negative sign implies that increasing the photon energy decreases the wavelength. We are only interested in changes or spreads, thus $\Delta\lambda/\Delta(h\upsilon) \approx |d\lambda/d(h\upsilon)|$, and this spread should be around $\lambda = \lambda_o$, so that Eq. (3.11.4) gives

$$\Delta\lambda = \frac{\lambda_o^2}{hc}\Delta(h\upsilon) = \lambda_o^2 \frac{3kT}{hc}$$

where we used $\Delta(h\nu) = 3kT$. We can substitute $\lambda = 1310$ nm and $T = 300$ K to calculate the linewidth of the 1310 nm LED

$$\Delta\lambda = \lambda^2 \frac{3kT}{hc} = (1310 \times 10^{-9})^2 \frac{3(1.38 \times 10^{-23})(300)}{(6.626 \times 10^{-34})(3 \times 10^8)} = 1.07 \times 10^{-7}\,\text{m} \quad \text{or} \quad 107\,\text{nm}$$

The spectral linewidth of an LED output is due to the spread in the photon energies, which is fundamentally about $3kT$. The only option for decreasing $\Delta\lambda$ at a given wavelength is to reduce the temperature. The output spectrum of a laser, on the other hand, has a much narrower linewidth.

EXAMPLE 3.11.2 LED spectral width

Consider the three experimental points in Figure 3.32 (b) as a function of temperature. By a suitable plot find m and verify Eq. (3.11.3).

Solution

From Example 3.11.1, we can use Eq. (3.11.3) with m instead of 3 as follows

LED linewidth and temperature

$$\frac{\Delta\lambda}{\lambda_o^2} = \left(\frac{mk}{hc}\right)T \tag{3.11.5}$$

and plot $\Delta\lambda/\lambda_o^2$ vs. T. The slope of the best line forced through zero should give mk/hc and hence m. Using the three λ_o and $\Delta\lambda$ values in the inset of Figure 3.32 (b), we obtain the graph in Figure 3.34. The best line is forced through zero to follow Eq. (3.11.5), and gives a slope of $1.95 \times 10^{-7}\,\text{nm}^{-1}\,\text{K}^{-1}$ or $195\,\text{m}^{-1}\,\text{K}^{-1}$. Thus,

$$\text{Slope} = 195\ \text{m K}^{-1} = \frac{m(1.38 \times 10^{-23}\ \text{J K}^{-1})}{(6.626 \times 10^{-34}\ \text{J s})(3 \times 10^8\ \text{m s}^{-1})}$$

so that

$$m = 2.81$$

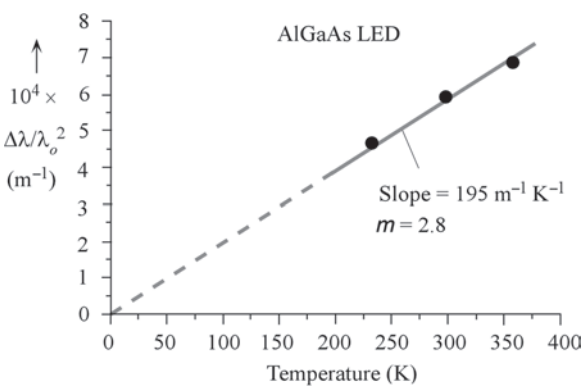
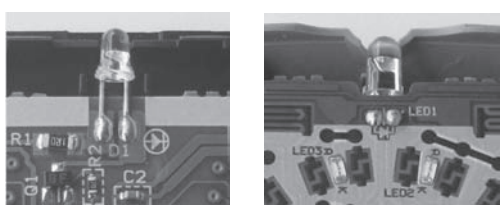


FIGURE 3.34 The plot of $\Delta\lambda/\lambda_o^2$ vs. T for an AlGaAs infrared LED, using the peak wavelength λ_o and spectral width $\Delta\lambda$ at three different temperatures, using the data shown in Figure 3.32 (b).



Infrared LEDs are widely used in various remote controls.

EXAMPLE 3.11.3 Dependence of the emission peak and linewidth on temperature

Using the Varshni equation, Eq. (3.11.2), find the shift in the peak wavelength (λ_o) emitted from a GaAs LED when it is cooled from 25°C to −25°C. The Varshni constants for GaAs are $E_{go} = 1.519\,\text{eV}$, $A = 5.41 \times 10^{-4}\,\text{eV K}^{-1}$, $B = 204\,\text{K}$.

Solution

At $T = 298\,\text{K}$, using the Varshni equation

$$E_g = E_{go} - AT^2/(B + T)$$
$$= 1.519\,\text{eV} - (5.41 \times 10^{-4}\,\text{eV K}^{-1})(298\,\text{K})^2/(204\,\text{K} + 298\,\text{K}) = 1.423\,\text{eV}$$

At 298 K, $(1/2)k_BT = 0.0128$ eV. The peak emission is at $h\upsilon_o \approx E_g + (1/2)k_BT$. Using $\upsilon_o = c/\lambda_o$, we get

$$\lambda_o = \frac{ch}{(E_g + \frac{1}{2}k_BT)} = \frac{(3 \times 10^8\,\text{m s}^{-1})(6.626 \times 10^{-34}\,\text{J s})/(1.602 \times 10^{-19}\,\text{eV J}^{-1})}{(1.4223\,\text{eV} + 0.0128\,\text{eV})} = 864.2\,\text{nm}$$

At −25°C, or 248 K, $(1/2)\,k_BT = 0.0107$ eV, repeating the above calculation

$$E_g = 1.519\,\text{eV} - (5.41 \times 10^{-4}\,\text{eV K}^{-1})(248\,\text{K})^2/(204\,\text{K} + 248\,\text{K}) = 1.445\,\text{eV}$$

and the new peak emission wavelength λ_o' is

$$\lambda_o' = \frac{(3 \times 10^8\,\text{m s}^{-1})(6.626 \times 10^{-34}\,\text{J s})/(1.602 \times 10^{-19}\,\text{eV J}^{-1})}{(1.445\,\text{eV} + 0.0107\,\text{eV})} = 852.4\,\text{nm}$$

The change $\Delta\lambda = \lambda_o - \lambda_o' = 864.2 - 852.4 = 11.8\,\text{nm}$ over 50°C, or 0.24 nm/°C. The examination of Figure 3.32 (b) shows that the change in the peak wavelength per unit temperature in the range −40°C to 85°C is roughly the same. Because of the small change, we kept four significant figures in E_g and λ_o calculations.

3.12 QUANTUM WELL HIGH INTENSITY LEDS

A typical **quantum well** (QW) device has an ultra-thin, typically less than 50 nm, narrow bandgap semiconductor with a bandgap E_{g1} sandwiched between two wider bandgap semiconductors with a bandgap E_{g2}, as illustrated in Figure 3.35 (a). For example, this could be a thin GaAs (E_{g1}) layer sandwiched between two $Al_xGa_{1-x}As$ (E_{g2}) layers. The wide bandgap layers are called **confining** layers. We assume that the two semiconductors are lattice matched in the sense that they have the same crystal structure and lattice parameter *a.* This means that interface defects due to the mismatch of crystal dimensions between the two semiconductor crystals are minimal; and neglected. Since the bandgap, E_g, changes at the interface, there are discontinuities in E_c and E_v at the interfaces as before; these discontinuities, ΔE_c and ΔE_v, are shown in Figure 3.35 (b) and depend on the semiconductor properties.[21] Because of the potential energy barrier, ΔE_c, conduction electrons in the thin E_{g1}-layer are confined in the *x*-direction. This confinement length *d*, the width of the thin E_{g1}-semiconductor, is so small that we can treat the electron as

[21] In the case of GaAs/AlGaAs heterostructure, ΔE_c is greater than ΔE_v. Very approximately, the change from the wider E_{g2} to narrower E_{g1} is proportioned 60% to ΔE_c and 40% to ΔE_v.

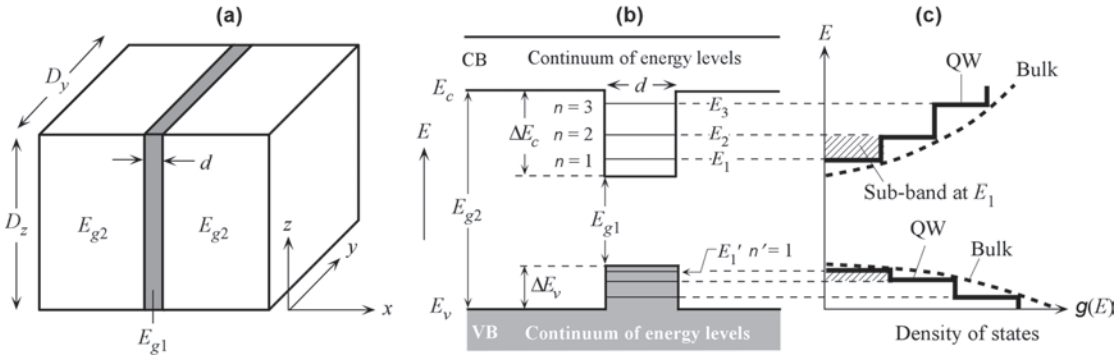


FIGURE 3.35 (a) A single quantum well (SQW) of a smaller bandgap material (E_{g1}) of thickness d along x surrounded by a thicker material of wider bandgap (E_{g2}). (b) The electron energy levels associated with motion along x are quantized as E_1, E_2, E_3, *etc.* Each level is characterized by a quantum number n. (c) The density of states for a bulk semiconductor and a QW.

in a one-dimensional (1D) potential energy (PE) well in the x-direction but as if it were free in the yz plane.

The energy of the electron in the QW must reflect its 1D quantization in the x-direction, and its freedom in the yz plane. If E_n is the electron energy in the well, then

Energy in a 1D quantum well

$$E_n = E_c + \frac{h^2 n^2}{8m_e^* d^2} + \frac{\hbar^2 k_y^2}{2m_e^*} + \frac{\hbar^2 k_z^2}{2m_e^*} \qquad \textbf{(3.12.1)}$$

where n is a quantum number having the values 1, 2, 3,…, and k_y and k_z are the wave vectors of the electron along y- and z-directions. The reason for the E_c in Eq. (3.12.1) is that the potential energy barriers are defined with respect to E_c. These PE barriers are ΔE_c along x and electron affinity (energy required to take the electron from E_c to vacuum) along y and z. The second term is the energy of an electron in an infinite PE well, whereas we have a finite PE well of depth ΔE_c. Thus, the second term is only an approximation. The minimum energy E_1 corresponds to $n = 1$ and is above E_c of the E_{g1}-semiconductor as shown in Figure 3.35 (b). For any given n value, we have a sub-band of energies due to k_y and k_z terms in Eq. (3.12.1). The separation between the energy levels associated with motion in the yz plane in a sub-band is so small that the electron is free to move in the yz plane as if it were in the bulk semiconductor; we assume a continuum of energy as shown in Figure 3.36. We therefore have a *two-dimensional electron gas* which is confined in the x-direction. The holes in the valence band are confined by the potential energy barrier ΔE_v (hole energy is in the opposite direction to electron energy) and behave similarly as illustrated in Figure 3.35 (b). They are characterized by the quantum number $n' = 1, 2$, *etc.* corresponding to the levels E_1', E_2', *etc.* [only E_1' is marked in Figure 3.35 (b)].

The density of electronic states for the two-dimensional electron system is not the same as that for the bulk semiconductor. For a given electron concentration n, the density of states $g(E)$, the number of quantum states per unit energy per unit volume, is constant and does not depend on the electron energy. The density of states for the confined electron and that in the bulk semiconductor are shown schematically in Figure 3.35 (c). $g(E)$ is constant at E_1 until E_2, where it increases as a step and remains constant until E_3, where again it increases as a step by the same amount and at every value of E_n. Density of states in the valence band behaves similarly as shown in Figure 3.35 (c). Since the electron is free in the yz plane within the well, its kinetic energy will increase parabolically and continuously with the wave vector k_{yz} in this plane as

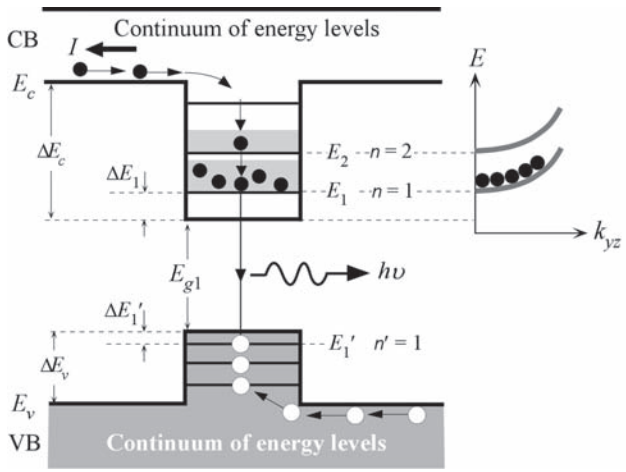


FIGURE 3.36 A QW structure that shows the energy levels in the wells and how charge carriers that are brought in by the current fall into the lowest energy level in the well and then recombine, emitting a photon. The electrons at a particular energy level also have kinetic energies in the *yz* plane, which is not quantized. The electrons are therefore spread in energy above E_n as shown. The same notion also applies to holes in the ΔE_v well.

shown in Figure 3.36. One has to account for this small kinetic energy, which is shown by shading the region above E_1, E_2, *etc.*, in Figure 3.36.

Since at E_1 there is a finite and substantial density of states, the electrons in the conduction band do not have to spread as far in energy as in the bulk to find states. In the bulk semiconductor, on the other hand, the density of states at E_c is zero and increases slowly with energy (as $E^{1/2}$), which means that the electrons are spread more deeply into the conduction band in search for states. A large concentration of electrons can easily occur near E_1 whereas this is not the case in the bulk semiconductor. Similarly, the majority of holes in the valence band will be around E_1' since there are sufficient states at this energy. Under a forward bias, as indicated in Figure 3.36, electrons are injected into the conduction band of the E_{g1}-layer, which serves as the *active layer.* The injected electrons readily populate the ample number of states at E_1 which means that the electron concentration at E_1 increases rapidly with the current. The direct recombination rate, that is, the radiative recombination rate, which determines the rate of photon emission, depends on the product *np*, the concentration of electrons *and* holes. Thus, the radiative transitions occur much more readily in the quantum well for the same current as in the bulk device. The quantum mechanical radiative recombination probability is also enhanced because the electrons and holes in the well are brought into close proximity. (However, there are still certain selection rules as there are for photon emission in the H-atom.) Furthermore, since the majority of electrons are spread around E_1 and holes around E_1', the range of emitted photon energies can be narrower than the emission spectrum in the corresponding bulk device; unless the well has been flooded with electrons at large currents as explained below. The loss of energy as the electron drops from $n = 3$ to 2 to 1, *etc.*, to lower levels inside the well occurs by the emission of phonons (lattice vibrations) and is very rapid. The transition from $n = 1$ to $n' = 1$ emits a photon of energy $h\upsilon \approx E_1 - E_1'$. (An approximate sign is used since the electron and the hole can have kinetic energies in the *yz* plane). The radiative transition in a QW must obey a **selection rule**, which requires the initial and final quantum numbers, n and n', to be the same.[22] The transition from $n = 1$ to $n' = 1$ is allowed and emits a photon, as well as that from $n = 2$ to $n' = 2$.

[22] Quantum mechanical selection rules for radiative transitions are nothing unusual; they also occur in radiative transitions in atoms. A very readable derivation can be found in M. Fox, *Optical Properties of Solids,* 2nd Edition (Oxford University Press, 2010), Ch. 6.

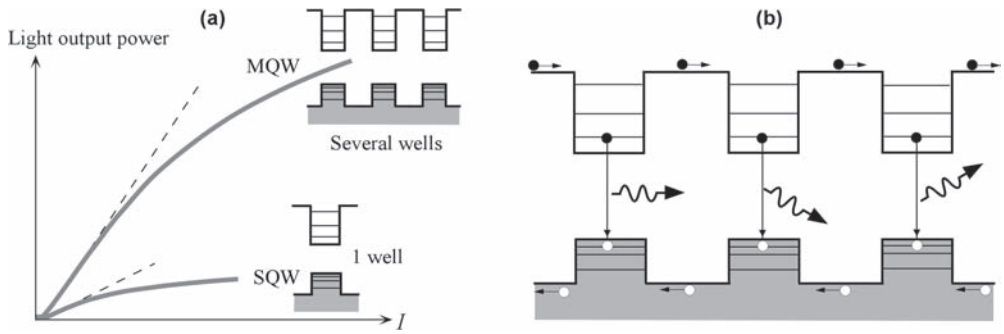


FIGURE 3.37 A schematic illustration of the comparison of light power output vs. current characteristics for an SQW and an MQW LED.

The main problem with the single quantum well (SQW) heterostructure LEDs is that, under a sufficiently large current, the well can be flooded with charge carriers and can overflow. For example, electrons can flood the QW ΔE_c and the well will overflow. The advantages of the QW action (such as confinement that increases the electron concentration n) would be lost. The light output will no longer increase proportionally to the current, and will fall behind the increase in the current as indicated in Figure 3.37 (a). This problem can be resolved by using **multiple quantum wells** (MQWs), in which electrons are shared by a number of quantum wells as in Figure 3.37 (b). Not only is the overflow problem alleviated but more photon flux is generated due to more wells as shown in Figure 3.37 (a). Modern high intensity UV, violet, and blue LEDs use MQW heterostructures. They use a thin $In_xGa_{1-x}N$ (E_{g1}) QW layer that is sandwiched between GaN (E_{g2}) layers. GaN has a large bandgap of 3.4 eV, and the composition and hence the bandgap of InGaN is chosen for the application, for example, for blue, $E_{g1} = 2.7\,\text{eV}$. The heterostructure has a number of MQWs to improve the efficiency but the number of QWs is not many, which is limited by the fabrication process.

EXAMPLE 3.12.1 Energy levels in the quantum well

Consider a GaAs QW sandwiched between two $Al_{0.40}Ga_{0.60}As$ layers. Suppose that the barrier height ΔE_c is 0.30 eV, the electron effective mass in the well is $0.067m_e$, and the width of the QW (d) is 12 nm. Calculate the energy levels E_1 and E_2 from the bottom of the well (E_c) assuming an infinite PE well as in Eq. (3.12.1). Compare these with the calculations for a *finite* PE well that give 0.022 eV, 0.088, and 0.186 for $n = 1$, 2, and 3.

Solution

We use Eq. (3.12.1) with $m_e^* = 0.067m_e$, $d = 12 \times 10^{-9}$ nm, so that for $n = 1$

$$\Delta E_n = E_n - E_c = \frac{h^2n^2}{8m_e^*d^2} = \frac{(6.624 \times 10^{-34}\,\text{J s})^2(1)^2/(1.602 \times 10^{-19}\,\text{J eV}^{-1})}{8(0.067 \times 9.1 \times 10^{-31}\,\text{kg})(12 \times 10^{-9}\,\text{m})^2} = 0.039\,\text{eV}$$

We can repeat the above calculation for $n = 2$ and 3 to find $\Delta E_2 = 0.156$ eV and $\Delta E_3 = 0.351$ eV. The third level will be above the well depth ($\Delta E_c = 0.3\,\text{eV}$). Clearly, the infinite QW predicts higher energy levels, by a factor of 1.8, and puts the third level inside the well, not outside. The finite QW calculation is not simple, and involves a numerical solution.[23]

[23] A similar example is solved for the finite well in M. Fox, *Optical Properties of Solids*, 2nd Edition (Oxford University Press, 2010), pp. 149–152. (This is a highly recommended text on optical properties.)

3.13 LED MATERIALS AND STRUCTURES

A. LED Materials

There are various direct bandgap semiconductor materials that can be readily doped to make commercial *pn* junction LEDs which emit radiation in the red and infrared range of wavelengths. The fabrication of an actual LED would require that the doped crystal layers with the required bandgap can be grown on a suitable **substrate** crystal. The substrate crystal and the LED material will have to be **lattice matched**, that is, they must have the same crystal structure and very close lattice parameters (a) to avoid creating dislocations at the interface. (Dislocations and other defects provide a nonradiative recombination path.) Lattice matching is not always perfect and often dictates what material compositions can be used.

An important class of commercial semiconductor materials that cover the visible spectrum is the **III–V ternary alloys** based on alloying GaAs and GaP, which are denoted as $GaAs_{1-y}P_y$. In this compound, As and P atoms from group V are distributed randomly at normal As sites in the GaAs crystal structure. When $y < 0.45$, the alloy $GaAs_{1-y}P_y$ is a direct bandgap semiconductor and hence the EHP recombination process is direct and efficient. The emitted wavelengths range from about 630 nm, red, for $y = 0.45$ ($GaAs_{0.55}P_{0.45}$) to 870 nm for $y = 0$, GaAs.

$GaAs_{1-y}P_y$ alloys (which includes GaP) with $y > 0.45$ are indirect bandgap semiconductors. The EHP recombination processes occur through recombination centers and involve lattice vibrations rather than photon emission. As mentioned above, however, if we add isoelectronic impurities such as nitrogen into the semiconductor crystal then the N-dopants can act as recombination centers. An electron is first captured by the N-center, the excess energy is lost to phonons, and then while at the N-center, it recombines with a hole in a radiative transition. The emitted photon energy is only slightly less than E_g. Nitrogen-doped indirect bandgap $GaAs_{1-y}P_y$ alloys are widely used in inexpensive green, yellow, and orange LEDs.

There are various commercially important direct bandgap semiconductor materials that emit in the red and infrared wavelengths which are typically **ternary** (containing three elements) and **quaternary** (four elements) alloys based on Group III and V elements, so-called **III–V alloys**. For example, GaAs with a bandgap of about 1.42 eV emits radiation at around 870 nm in the infrared. But ternary alloys based on $Al_{1-x}Ga_xAs$ where $x < 0.43$ are direct bandgap semiconductors. The composition can be varied to adjust the bandgap and hence the emitted radiation from about 640–870 nm, from deep red light to infrared.

AlGaInP is a quaternary III–V alloy (In, Ga, Al from III, and P from V) that has a direct bandgap variation with composition over the visible range. It can be lattice-matched to GaAs substrates for compositions $(Al_xGa_{1-x})_{0.5}In_{0.5}P$ where $x < 0.53$, that is, $Ga_{0.50}In_{0.50}P$ ($E_g = 1.89$ eV, red) to $Al_{0.265}Ga_{0.235}In_{0.50}P$ (2.33 eV, green). Many brands of high-intensity LEDs have been based on this material, which is likely to continue to be used in the high-intensity visible LED range, especially for the red, amber, and yellow.

The bandgap of quaternary alloys $In_{1-x}Ga_xAs_{1-y}P_y$ can be varied with composition (x and y) to span wavelengths from 870 nm (GaAs) to 3.5 μm (InAs) which includes the optical communication wavelengths of 1.3 μm and 1.55 μm. Figure 3.38 summarizes some typical wavelengths that can be emitted for a few selected semiconductor materials over the range from 0.3 μm to 1.7 μm, or from the UV to IR.

GaN is a direct bandgap semiconductor with an E_g of 3.4 eV. The blue GaN LEDs actually use the GaN alloy InGaN with a bandgap of about 2.7 eV, which corresponds to blue emission. One of the most important technological advances in the last two decades has been the development of various III-Nitride LEDs that can emit with high intensities from the UV to green. The alloys

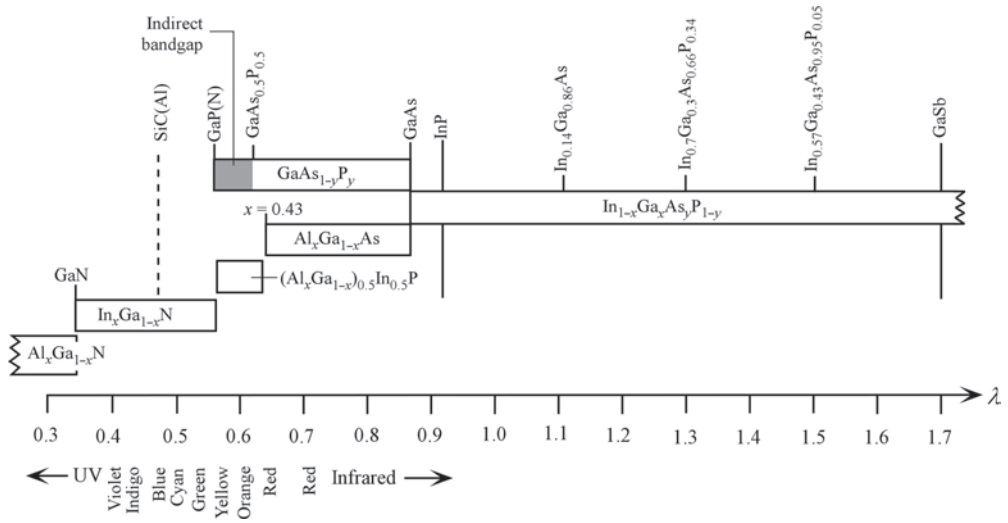


FIGURE 3.38 Free-space wavelength coverage by different LED materials from the visible spectrum to the infrared, including wavelengths used in optical communications. Grey region and dashed lines are indirect E_g materials. Only material compositions of importance have been shown.

of GaN ($E_g = 3.4\,\text{eV}$) with InN ($E_g = 0.77\,\text{eV}$), $In_xGa_{1-x}N$ spans wavelengths from the UV up to the IR, though they are currently not used beyond the green wavelength as other semiconductors such as AlGaInP provide better efficiencies. The alloys of AlN ($E_g = 6.2$ eV) and GaN ($E_g = 3.4$ eV), AlGaN, provide for wavelengths in the UV. GaN can be doped *n*-type (*e.g.*, with Si or Ge) and *p*-type (*e.g.*, with Mg), and the GaN LEDs are generally MQW heterostructures.

B. LED Structures

In its simplest technological form, LEDs are typically fabricated by *epitaxially* growing doped semiconductor layers on a suitable substrate (*e.g.*, GaAs or GaP) as illustrated for a simple homojunction *pn* junction LED in Figure 3.39 (a). This type of planar *pn* junction is formed by the epitaxial growth of first the *n*-layer and then the *p*-layer. The substrate is essentially a mechanical support for the *pn* junction device (the layers) and can be of different crystal. The *p*-side is on the surface from which light is emitted and is therefore made narrow (a few microns)

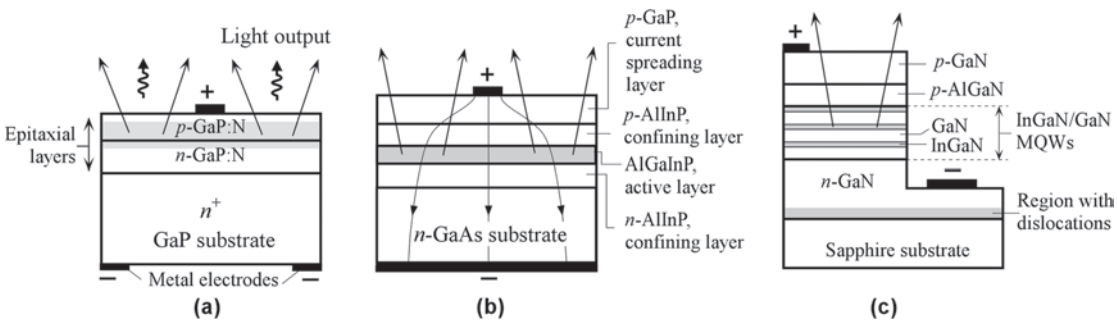


FIGURE 3.39 A schematic illustration of various typical LED structures. (a) A planar surface-emitting homojunction green GaP:N LED. (b) AlGaInP high-intensity heterostructure LED. (c) III-Nitride-based (GaN/InGaN) MQW LED for emission from the UV to green.

to allow the photons to escape without being reabsorbed. To ensure that most of the recombination takes place in the *p*-side, the *n*-side is heavily doped (n^+). Those photons that are emitted toward the *n*-side become either absorbed or reflected back at the substrate interface depending on the substrate thickness and the exact structure of the LED. The use of a segmented back electrode as in Figure 3.39 (a) will encourage reflections from the semiconductor–air interface.

If the epitaxial layer and the substrate crystals have different crystal lattice parameters, then there is a lattice mismatch between the two crystal structures. This causes lattice strain in the LED layer and hence leads to crystal defects. Such crystal defects encourage radiationless EHP recombinations. That is, a defect acts as a recombination center. Such defects are reduced by lattice matching the LED epitaxial layer to the substrate crystal. It is therefore important to lattice-match the LED layers to the substrate crystal. For example, $Al_xGa_{1-x}As$ alloys with $x < 0.43$ are direct bandgap semiconductors that have a bandgap corresponding to the IR to the red emission region (Figure 3.38). $Al_xGa_{1-x}As$ layers can be grown on GaAs substrates with excellent lattice match, which results in high-efficiency LED devices.

Figure 3.39 (b) shows the structure of a high-intensity AlGaInP heterostructure LED. The layers are grown epitaxially on an *n*-GaAs substrate. There are at least four layers. The active layer is a thin AlGaInP (*e.g.*, $Al_{0.35}Ga_{0.15}In_{0.5}P$), which is lightly doped. This layer is sandwiched by confining layers that are *p*-type and *n*-type AlInP (*e.g.*, $Al_{0.5}In_{0.5}P$) on the positive and negative terminal sides, respectively. AlInP has a wider bandgap than AlGaInP, and the band offsets confine the carriers to the active region. Remember that under forward bias the *p*-AlInP injects holes and *n*-AlInP injects electrons into the active layer. The top layer is *p*-GaP and serves to spread out the current to regions outside the top contact. Thus, radiative recombinations are avoided right under the top contact from which photons cannot be extracted. Put differently, EHP recombination is spread into regions outside the zone directly under the top contact.

Figure 3.39 (c) shows a simplified III-Nitride-based MQW LED that can be used for emission in the blue as well as green.[24] With some modification to compositions, it can also emit in the UV. The *p*-GaN (doped with Mg) is the required *p*-layer used for the injection of holes. The QWs are formed between the narrower E_g InGaN and wider E_g GaN as described above, which are undoped. There is a *p*-AlGaN layer that is called a *buffer* layer. The bandgap of AlGaN is wider than InGaAs, so it confines the injected electrons in the QW-region. The *n*-GaN layer is the electron injecting *n*-type semiconductor from which electrons are injected into the MQWs. It is difficult to find a matching crystal substrate for GaN. A sapphire crystal is the most commonly used substrate, though the mismatch is roughly 12% (significant). Special growth techniques have been developed to keep the defects (dislocations) to the initial GaN growth region near the sapphire–GaN interface, away from actual LED heterostructure. Notice that the negative terminal is on high-quality *n*-GaN, away from the defective region.

Not all light rays reaching the semiconductor–air interface, however, can escape because of total internal reflection (TIR). Those rays with angles of incidence greater than the critical angle θ_c become reflected as illustrated in Figure 3.40 (a). For the GaAs–air interface, for example, θ_c is only 17°, which means that much of the light suffers TIR. It is possible to shape the surface of the semiconductor into a dome, or hemisphere, so that light rays strike the surface at angles less than θ_c and therefore do not experience TIR. The main drawback, however, is the additional difficult

[24] III-Nitride MQW LEDs are currently the most efficient commercial LEDs in the blue, violet, and the UV. An extensive review can be found in Hadis Morkoç, *Handbook of Nitride Semiconductors and Devices*, Vol. 3 (Wiley-VCH, Weinheim, 2008), Ch. 1.

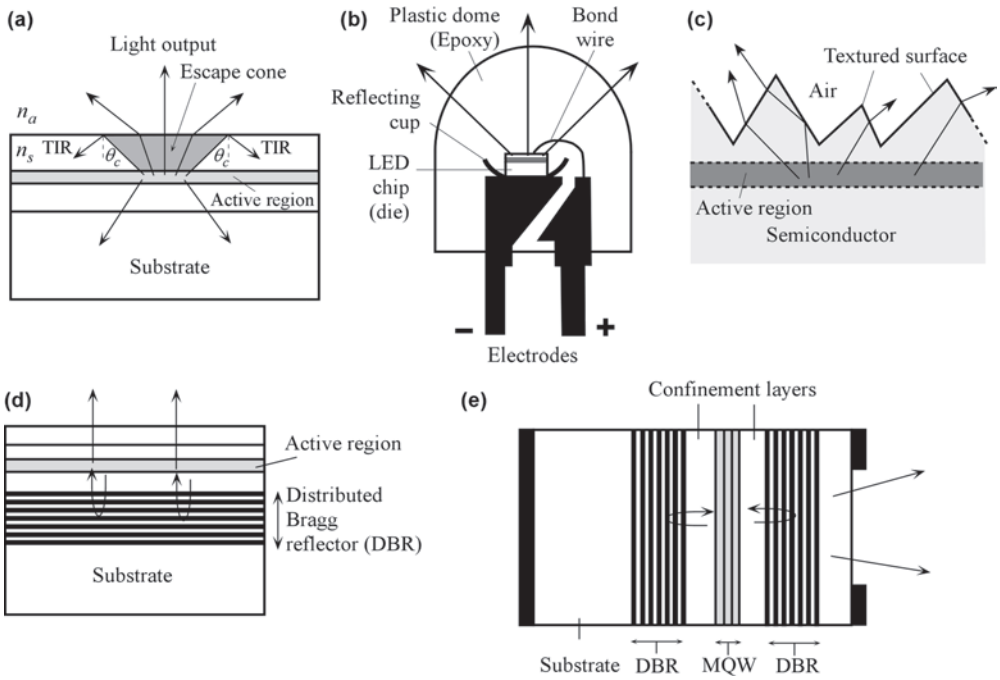


FIGURE 3.40 (a) Some of the internally generated light suffers total internal reflection (TIR) at the semiconductor–air interface and cannot be emitted into the outside. (b) A simple structure that overcomes the TIR problem by placing the LED chip at the centre of a hemispherical plastic dome. The epoxy is refractive index matched to the semiconductor and the rays reaching the dome's surface do not suffer TIR. (c) An example of a textured surface that allows light to escape after a couple of (or more) reflections (highly exaggerated sketch). (d) A distributed Bragg reflector (DBR) under the confining layer (below the active region in grey) acts as a dielectric mirror, and increases the extraction ratio. (e) An RCLED is an LED with an optical resonant cavity (RC) formed by two DBRs has a narrower emission spectrum.

process in fabricating such domed LEDs and the associated increase in expense. An inexpensive and common procedure that reduces TIR is the encapsulation of the semiconductor junction within a transparent plastic medium (an epoxy) which has a higher refractive index than air and, further, also has a domed surface on the emission side of the LED chip as shown in Figure 3.40 (b). Many individual LEDs are sold in similar types of plastic bodies.



Shuji Nakamura obtained his PhD from the University of Tokushima in Japan, and is currently a professor at the University of California at Santa Barbara and the director of Solid State Lighting and Energy Center. He has been credited with the pioneering work that has led to the development of GaN and $In_xGa_{1-x}N$-based blue and violet light–emitting diodes and laser diodes. He discovered how III-Nitrides could be doped *p*-type, which opened the way to fabricating various UV, violet, blue, and green LEDs. He is holding a blue laser diode that is turned on. *(Courtesy of Shuji Nakamura, University of California, Santa Barbara.)*

Another example of a device structure that improves the light extraction ratio is shown in Figure 3.40 (c). The surface has been textured or nanostructured. Such a textured surface allows light to escape after one or two reflections. The extraction ratio can be greatly improved if the rays traveling toward the substrate could be somehow reflected back toward the exit window. In some LEDs, a *distributed Bragg reflector* (DBR), that is a dielectric mirror, under the confining layer (below the active region) acts as a dielectric mirror at the wavelength of the LED light, and increases the extraction ratio, as shown in Figure 3.40 (d).

Another possible design is to use a *resonant cavity* for the light generated by the active region as shown in Figure 3.40 (e). An RCLED (resonant cavity LED) is an LED with an optical resonant cavity (RC) formed by two DBRs. The optical cavity is obviously wavelength-selective since only those special modes of the optical cavity that fall into the spontaneous emission spectrum can be supported or excited. The width of the emitted spectrum is significantly narrower, as much as ten times, than a conventional LED. The reason is that the light from the active region can only escape if it is a mode of the resonant cavity. The spectral width of this light is determined by the resonant cavity losses rather than the emission spectrum from the active region.

EXAMPLE 3.13.1 Light extraction from a bare LED chip

As shown in Figure 3.40 (a), due to total internal reflection (TIR) at the semiconductor–air surface, only a fraction of the emitted light can escape from the chip. The critical angle θ_c is determined by $\sin\theta_c = n_a/n_s$ where n_a and n_s are the refractive indices of the ambient (*e.g.,* for air, $n_a = 1$) and the semiconductor, respectively. The light within the **escape cone** defined by θ_c can escape into the ambient without TIR as indicated in Figure 3.40 (a). To find the fraction of light within the escape cone we need to consider solid angles, which leads to $(1/2)(1 - \cos\theta_c)$. Further, suppose that T is the average light *transmittance* of the n_s–n_a interface for those rays within the escape cone, then for a simple bare chip,[25]

$$\text{Light extraction ratio} \approx (1/2)(1 - \cos\theta_c) \times T \qquad \textbf{(3.13.1)}$$

Light extraction ratio

Estimate the extraction ratio for a GaAs chip with $n_s = 3.4$ and air as ambient ($n_a = 1$) and then with an epoxy dome with $n_a = 1.8$.

Solution

First, note that $\theta_c = \arcsin(n_a/n_s) = \arcsin(1/3.4) = 17.1°$. For T we will assume near-normal incidence (somewhat justified since the angle 17.1° is not too large) so that from Chapter 1,

$$T = 4n_s n_a/(n_s + n_a)^2 = 4(3.4)(1)/(3.4 + 1)^2 = 0.702$$

Using Eq. (3.13.1)

$$\text{Light extraction ratio} \approx (1/2)(1 - \cos\theta_c) \times T = (1/2)\left[1 - \cos(17.1°)\right] \times 0.702$$

$$\approx 0.0155 \quad \text{or} \quad 1.6\%$$

It is clear that only 1.6% of the generated light power is extracted from a bare chip, which is disappointingly small. The technological drive is therefore to improve light extraction as much as possible. If we now repeat the calculation for $n_a = 1.8$, we would find, $\theta_c = 32°$, and 6.9% light extraction.

[25] An extensive treatise on various LED structures and light extraction techniques can be found in E. F. Schubert, *Light-Emitting Diodes*, 2nd Edition (Cambridge University Press, 2006). Equation (3.13.1) is an approximation inasmuch as we should average the transmittance.

3.14 LED EFFICIENCIES AND LUMINOUS FLUX

The comparison of different LED materials and device structures requires that we define internal and external quantum efficiency. Further, we also need to meaningfully compare the output power and brightness of different light-emitting devices per unit electrical energy input.

The **internal quantum efficiency** (IQE) η_{IQE} gauges what fraction of electron hole recombinations in the forward-biased *pn* junction are radiative and therefore lead to photon emission. Nonradiative transitions are those in which an electron and a hole recombine through a recombination center such as a crystal defect or an impurity and emit phonons (lattice vibrations). Suppose that τ_r is the mean lifetime of a minority carrier before it recombines radiatively and τ_{nr} is the mean lifetime before it recombines via a recombination center (or a defect) without emitting a photon. By definition,

Internal quantum efficiency

$$\eta_{\text{IQE}} = \frac{\text{Rate of radiative recombination}}{\text{Total rate of recombination (radiative and nonradiative)}} \tag{3.14.1}$$

or

Internal quantum efficiency

$$\eta_{\text{IQE}} = \frac{\tau_r^{-1}}{\tau_r^{-1} + \tau_{nr}^{-1}} \tag{3.14.2}$$

The rate of nonradiative recombination rate $(1/\tau_{nr})$ would also include the recombination of injected carriers at the interfaces of heterostructures between different crystals: an important loss mechanism that would reduce η_{IQE}.

The total current I is determined by the total rate of recombinations whereas the number of photons emitted per second, the **photon flux** Φ_{ph}, is determined by the rate of radiative recombinations.

Internal quantum efficiency

$$\eta_{\text{IQE}} = \frac{\text{Photons emitted per second}}{\text{Total carriers lost per second}} = \frac{\Phi_{\text{ph}}}{I/e} = \frac{P_{o(\text{int})}/h\upsilon}{I/e} \tag{3.14.3}$$

where $P_{o(\text{int})}$ is the *optical* power generated internally (not yet extracted).

The **external quantum efficiency** (EQE) η_{EQE} of an LED represents the efficiency of conversion from electrical quanta, *i.e.,* electrons, that flow into the LED to optical quanta, *i.e.,* photons, that are emitted into the outside world. It incorporates the "internal" efficiency of the radiative recombination process [embedded in Eq. (3.14.1)] and the subsequent efficiency of photon extraction from the device. Suppose that the actual optical power emitted to the ambient, called the **radiant flux**, is P_o. (Φ_e is also used in the literature.) $P_o/h\upsilon$ is the number of emitted photons per second. Since the number of electrons flowing into the LED is I/e, we have

External quantum efficiency

$$\eta_{\text{EQE}} = \frac{P_o/h\upsilon}{I/e} \tag{3.14.4}$$

Thus, we should use external QE to meaningfully compare different LED efficiencies and internal QE in comparing different LED materials. For indirect bandgap semiconductors η_{EQE} are generally less than 1% whereas for direct bandgap semiconductors with the right device structure, η_{EQE} can be substantial, for example, 30–40%.

The light extraction ratio was introduced in Example 3.13.1. We can now define this more formally. The light **extraction ratio**, or the **extraction efficiency** (EE), η_{EE}, is the fraction of light that is extracted to the ambient from the internally generated light, that is,

$$\eta_{\text{EE}} = \frac{\text{Photons emitted externally from the device}}{\text{Photons generated internally by recombination}} \tag{3.14.5}$$

Extraction efficiency

Using Eqs. (3.14.1), (3.14.3) and (3.14.5), the emitted optical output power, the radiant flux, is

$$P_o = \eta_{\text{EE}} P_{o(\text{int})} = h\upsilon \eta_{\text{EE}} \eta_{\text{IQE}} (I/e) \tag{3.14.6}$$

Emitted optical power

The **power conversion efficiency** (PCE), η_{PCE}, or simply the **power efficiency**, gauges the overall efficiency of conversion from the input of electrical power to the output of optical power, *i.e.,*

$$\eta_{\text{PCE}} = \frac{\text{Optical output power}}{\text{Electrical input power}} = \frac{P_o}{IV} \approx \eta_{\text{EQE}} \left(\frac{E_g}{eV} \right) \tag{3.14.7}$$

Power efficiency

If we wish to compare the brightness of light-emitting devices per unit electrical power input then we need to examine the luminous flux emitted. The visual "brightness" of a source as observed by an average daylight-adapted eye is proportional to the radiation (optical) power emitted, that is, the radiant flux, and the efficiency of the eye to detect the spectrum of the emitted radiation. While the eye can see a red color source, it cannot see an infrared source and the brightness of the infrared source would be zero. The **luminous flux** Φ_v is a measure of *visual* **brightness**, in lumens (lm), and is defined by

$$\Phi_v = P_o \times (633 \text{ lm W}^{-1}) \times V(\lambda) \tag{3.14.8}$$

Luminous flux

where P_o is the radiant flux or the radiation power emitted (in watts) and $V(\lambda)$ is the **relative luminous efficiency** (or the relative sensitivity) of an average light-adapted (photopic) eye, which depends on the wavelength and hence λ in parenthesis. The function $V(\lambda)$ is also called the **luminosity function** and the **visibility function**. $V(\lambda)$ is a Gaussian-like function with a peak of unity at 555 nm as shown in Figure 3.41. One lumen of luminous flux, or brightness, is obtained from a 1.58 mW light source emitting at a single wavelength of 555 nm (green). A typical 60 W incandescent lamp provides roughly 900 lm (or 15 lm W^{-1}). When we buy a light bulb, we are buying lumens.

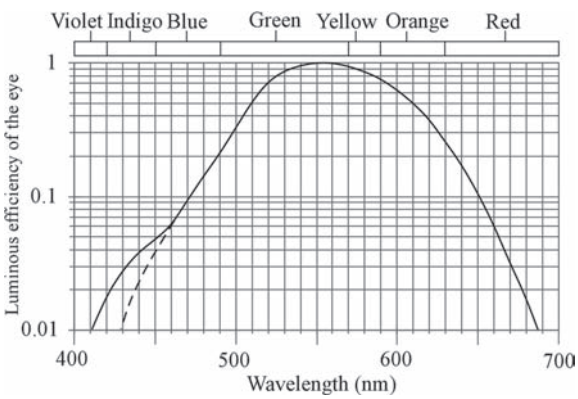


FIGURE 3.41 The luminous efficiency $V(\lambda)$ of the light-adapted (photopic) eye as a function of wavelength. The solid curve is the Judd-Vos modification of the CIE 1924 photopic photosensitivity curve of the eye. The dashed line shows the modified region of the original CIE 1924 curve to account for its deficiency in the blue-violet region. (The vertical axis is logarithmic.)

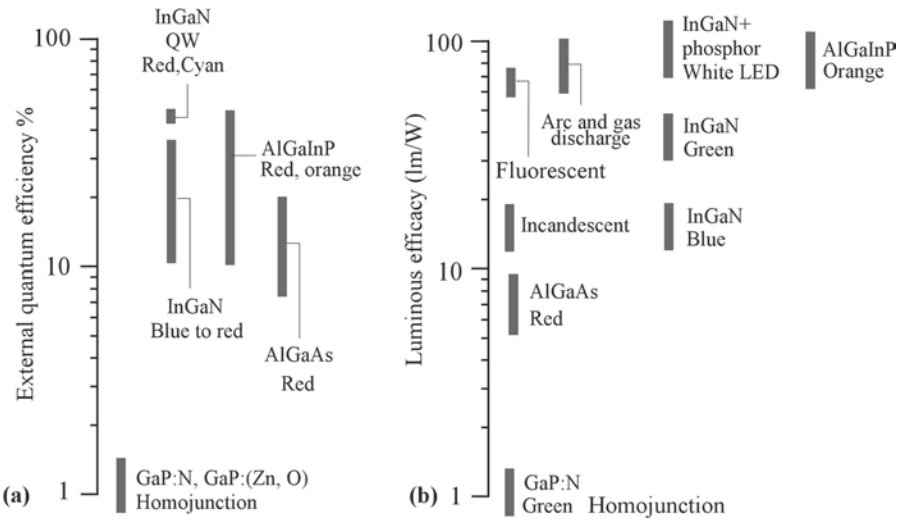


FIGURE 3.42 Typical (a) external quantum efficiency and (b) luminous efficacy of various selected LEDs, and how they stand against other light sources such as the fluorescent tube, arc, and gas discharge lamps and the incandescent lamp.

The **luminous efficacy**[26] **of a light source** (such as a lamp) as commonly used in *lighting applications* is the efficiency with which an electrical light source converts the input electric power (watts) into an emitted luminous flux (lumens).

Luminous efficacy (efficiency)

$$\eta_{\text{LE}} = \frac{\Phi_v}{IV} \tag{3.14.9}$$

A 100 W light bulb producing 1700 lumens has an efficacy of 17 lumens watt^{-1} (lm W^{-1}). Recent technological advances have led to LEDs with efficacies that are comparabale to standard fluorescent tubes—efficacies around 100 lm W^{-1}. LEDs as solid state lamps have much longer lifetimes and much higher reliability, and hence are expected to be more economical than incandescent and fluorescent lamps. Figure 3.42 (a) and (b) show some typical external efficiency and luminous efficacy values of various selected LEDs. The advantages of III-Nitrides (*e.g.*, InGaN) are clearly obvious. It is also important to note that in some cases, as in InGaN based LEDs, the high EQE and hence luminous efficacy cannot be maintained as the current is increased to obtain higher luminous flux (more lumens) because the output flux is not directly proportional to the input current, especially at high currents, as apparent in Figure 3.37 (a).

EXAMPLE 3.14.1 LED efficiencies

A particular 870 nm IR LED for use in optical links and instrumentation has a GaAs chip. Active layer that has been doped p-type with $2 \times 10^{17}\,\text{cm}^{-3}$ of acceptors and the nonradiative lifetime is about 100 ns. At a forward current of 30 mA, the voltage across it is 1.35 V, and the emitted optical power is 6.5 mW. Calculate the IQE, EQE, and PCE, and estimate the light extraction ratio. For GaAs, $B \approx 2 \times 10^{-16}\,\text{m}^2\,\text{s}^{-1}$.

[26] Some authors use the term *luminous efficiency* but the latter, strictly, needs the output and input quantities to have the same units so that the efficiency can be expressed as a percentage, which is not the case here. *Efficacy* would be a better term.

Solution

The radiative lifetime $\tau_r = 1/BN_a = 1/\left[(2 \times 10^{-16}\,\mathrm{m^3\,s^{-1}})(2 \times 10^{23}\,\mathrm{m^{-23}})\right] = 2.5 \times 10^{-8}\,\mathrm{s}$ or 25 ns. IQE is,

$$\eta_{\mathrm{IQE}} = \frac{\tau_r^{-1}}{\tau_r^{-1} + \tau_{nr}^{-1}} = \frac{(25\,\mathrm{ns})^{-1}}{(25\,\mathrm{ns})^{-1} + (100\,\mathrm{ns})^{-1}} = 0.80 \quad \text{or} \quad 80\%$$

The emitted photon energy $h\upsilon = hc/\lambda \approx 1.43$ eV. Thus the EQE is

$$\begin{aligned}\eta_{\mathrm{EQE}} &= (P_o/h\upsilon)/(I/e)\\ &= \left[(6.5 \times 10^{-3}\,\mathrm{W})/(1.43\,\mathrm{eV} \times 1.6 \times 10^{-19}\,\mathrm{J\,eV^{-1}})\right]/\left[(30 \times 10^{-3}\,\mathrm{A})/(1.6 \times 10^{-19}\,\mathrm{C})\right]\\ &= 0.15 \quad \text{or} \quad 15\%\end{aligned}$$

The PCE is simply P_o/IV or $(6.5\,\mathrm{mW})/\left[(30\,\mathrm{mA})(1.35\,\mathrm{V})\right]$, that is, 0.16, *i.e.*, 16%.
From Eq. (3.14.6), using $P_o = h\upsilon\eta_{\mathrm{EE}}\eta_{\mathrm{IQE}}(I/e)$,

$$6.5 \times 10^{-3}\,\mathrm{W} = (1.43\,\mathrm{eV} \times 1.6 \times 10^{-19}\,\mathrm{J\,eV^{-1}})\eta_{\mathrm{EE}}(0.80)(30 \times 10^{-3}\,\mathrm{A}/1.6 \times 10^{-19}\,\mathrm{C})$$

solving the above gives $\eta_{\mathrm{EE}} = 0.19$ or 19%.

EXAMPLE 3.14.2 LED brightness

Consider two LEDs, one red, with an optical output power (radiant flux) of 10 mW, emitting at 650 nm, and the other, a weaker 5 mW green LED, emitting at 532 nm. Find the luminous flux emitted by each LED.

Solution

For the red LED, at $\lambda = 650\,\mathrm{nm}$, Figure 3.41 gives $V \approx 0.10$ so that from Eq. (3.14.8)

$$\Phi_v = P_o \times (633\,\mathrm{lm\,W^{-1}}) \times V = (10 \times 10^{-3}\,\mathrm{W})(633\,\mathrm{lm\,W^{-1}})(0.10) = 0.63\,\mathrm{lm}$$

For the green LED, $\lambda = 532$ nm, Figure 3.41 gives $V \approx 0.87$ so that from Eq. (3.14.8)

$$\Phi_v = P_o \times (633\,\mathrm{lm\,W^{-1}}) \times V = (5 \times 10^{-3}\,\mathrm{W})(633\,\mathrm{lm\,W^{-1}})(0.87) = 2.8\,\mathrm{lm}$$

Clearly the green LED at half the optical power is 4 times brighter than the red LED.

3.15 BASIC LED CHARACTERISTICS

The *I–V* characteristics of LEDs depend on the device structure, including the material properties of the semiconductors used in the LED. The DC current generally increases steeply with the voltage, as shown in Figure 3.43 (a). The *I–V* characteristics do not always follow an exact exponential behavior that is often seen for a simple forward-biased *pn* junction. Most modern LEDs are heterostructure devices, with several layers of differently doped semiconductors. The current increases sharply over a narrow voltage range, as in Figure 3.43 (a), where it can be seen that there is an apparent **turn-on** or a **cut-in voltage** beyond which the current increases very sharply with voltage. Most LED manufacturers, however, quote the **forward voltage** V_F when the LED is operating fully; for example, the manufacturer of one type of red AlGaInP-based LED quotes $V_F = 2.0\,\mathrm{V}$ at $I = 20$ mA, at 40% rated maximum current. V_F depends on both the semiconductor material and the device structure. There is an overall trend in which V_F tends to increase with decreasing wavelength, that is, with increasing photon energy or roughly the bandgap E_g of the active region. However, the device structure is also very important, and can skew this

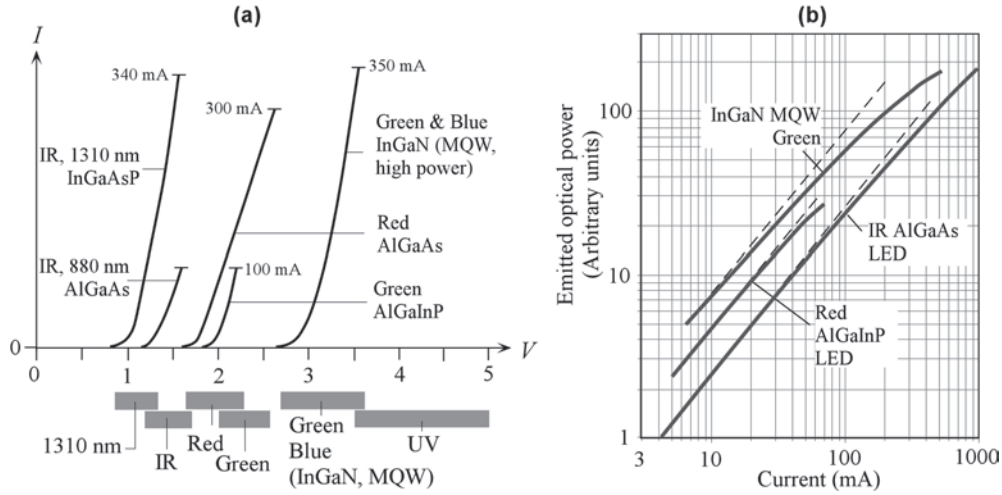


FIGURE 3.43 (a) Current-voltage characteristics of a few LEDs emitting at different wavelengths from the IR to blue. (b) Log-log plot of the emitted optical output power vs. DC current for three commercial devices emitting at IR (890 nm), red and green. The vertical scale is in arbitrary units and the curves have been shifted to show the dependence of P_o on I. The ideal linear behavior $P_o \propto I$ is also shown.

observation. MQW InGaN LEDs have a higher V_F than corresponding AlGaInP heterojunction LEDs, that is, they operate at a higher voltage across the device as apparent for the two green LEDs in Figure 3.43 (a). Indeed, InGaN green to violet LEDs all have very similar V_F.

Typical optical output power (P_o) vs. current (I) characteristics of LEDs are shown in Figure 3.43 (b) on a log-log plot for three cases. For comparison, the expected linear relationship, $P_o \propto I$, has been also shown for each device. In general, at high currents, P_o vs. I relationship curves down from the expected linear, $P_o \propto I$, behavior. The worst case is for InGaN MQW LEDs in which there is significant deviation from the expected linear relationship almost from the start, that is, P_o cannot keep up linearly with the current and droops as the current increases. This droop represents a drop in the quantum efficiency of the MQW InGaN device, and is currently a topical area of research. The P_o-I characteristics for standard AlGaInP and AlGaAs heterojunction LEDs deviate from linearity mainly at high currents, exhibiting an extensive range of reasonable linearly. While a non-linear P_o-I behavior is not a serious problem in digital communications, it can create distortion in analog modulation, especially under large signals.

3.16 LEDS FOR OPTICAL FIBER COMMUNICATIONS

The type of light source suitable for optical communications depends not only on the communication distance but also on the bandwidth requirement. For short-haul applications, for example, local networks, LEDs are preferred as they are simpler to drive, more economic, have a longer lifetime, and provide the necessary output power even though their output spectrum is much wider than that of a laser diode. LEDs are frequently used with graded index fibers inasmuch as typically dispersion in a graded index fiber is primarily due to intermodal dispersion rather than intramodal dispersion. (For example, at 1310 nm communications, the material and hence chromatic dispersion in a graded index fiber is usually much smaller than intermodal dispersion.) For long-haul and wide-bandwidth communications, invariably laser diodes are used because of their narrow linewidth, high output power, and higher signal bandwidth capability.

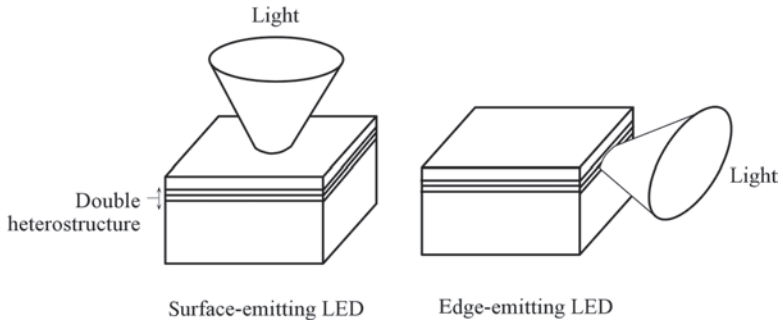


FIGURE 3.44
A surface-emitting and an edge-emitting LED.

There are essentially two types of LED devices which are illustrated in Figure 3.44. If the emitted radiation emerges from an area in the plane of the recombination layer as in (a) then the device is a **surface-emitting LED** (SLED). If the emitted radiation emerges from an area on an edge of the crystal as in (b), that is, from an area on a crystal face perpendicular to the active layer, then the LED is an **edge-emitting LED** (ELED).

The simplest method of coupling the radiation from a surface-emitting LED into an optical fiber is to etch a well in the planar LED structure and lower the fiber into the well as close as possible to the active region where emission occurs. This type of structure, as shown in Figure 3.45 (a), is called a **Burrus-**type device (after its originator). An epoxy resin is used to bond the fiber and provide refractive index matching between the glass fiber and the LED material to capture as much of the light rays as possible. Note that in the double heterostructure LED used in this way, the photons emitted from the active region, with a smaller bandgap, do not get absorbed by the neighboring layer, which has a wider bandgap. Another method is to use a truncated **spherical lens** (a microlens) with a high refractive index ($n = 1.9 - 2$) to focus the light into the fiber as shown in Figure 3.45 (b). The lens is bonded to the LED with a refractive index–matching cement and, in addition, the fiber can be bonded to the lens with a similar cement.

Edge-emitting LEDs provide a greater intensity light and also a beam that is more collimated than the surface-emitting LEDs. Figure 3.46 shows the structure of a typical edge-emitting LED for operation at ~1.5 μm. The light is guided to the edge of the crystal by a **dielectric waveguide** formed by wider bandgap semiconductors surrounding a double heterostructure. The recombination of injected carriers occurs in the InGaAs active region which has a bandgap $E_g \approx 0.83\,\text{eV}$. Recombination is confined to this layer because the surrounding InGaAsP layers, **confining layers**, have a wider bandgap ($E_g \approx 1\,\text{eV}$) and the InGaAsP/InGaAs/InGaAsP layers form a

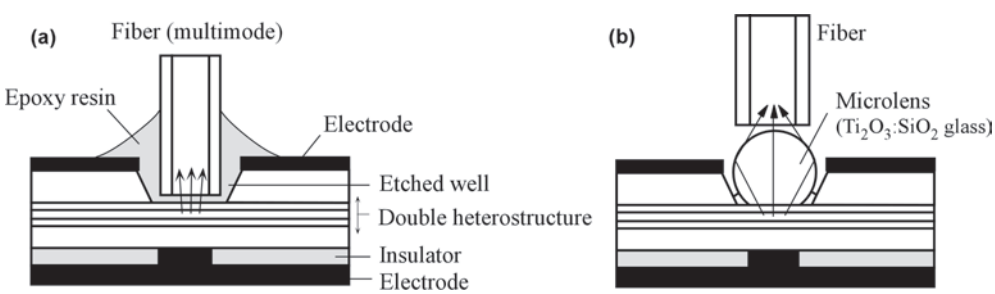


FIGURE 3.45 Coupling of light from surface-emitting LEDs into optical fibers. (a) Light is coupled from a surface-emitting LED into a multimode fiber using an index matching epoxy. The fiber is bonded to the LED structure. (b) A microlens focuses diverging light from a surface-emitting LED into a multimode optical fiber.

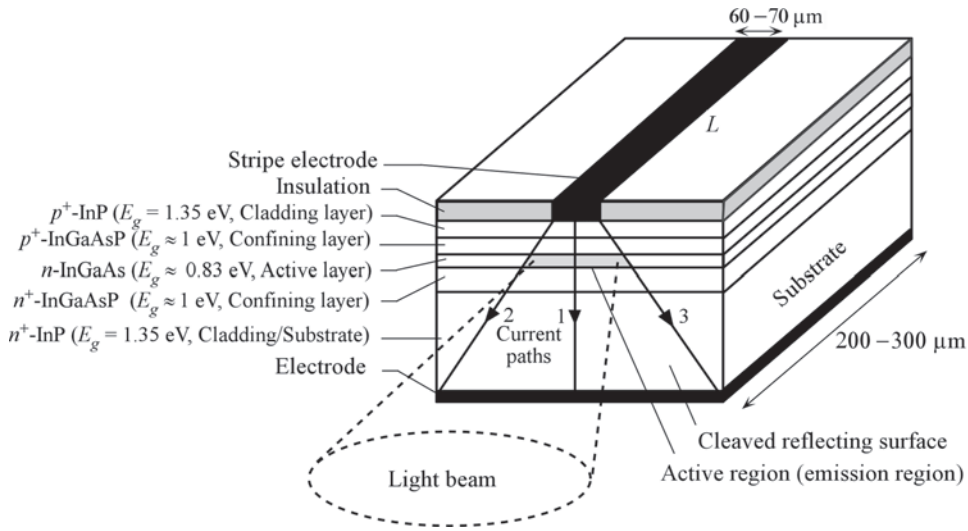


FIGURE 3.46 Schematic illustration of the structure of a double heterojunction stripe contact edge-emitting LED. (Upper case notation for a wider bandgap semiconductor is not used as there are several layers with different bandgaps.)

double heterostructure. The light emitted in the active region (InGaAs) spreads into the neighboring layers (InGaAsP) which guide it along the crystal to the edge. InP has a wider bandgap ($E_g \approx 1.35$ eV) and thus a lower refractive index than InGaAsP. The two InP layers adjoining the InGaAsP layers therefore act as **cladding layers** and thereby confine the light to the DH structure.

Generally some kind of lens system is used to conveniently couple the emitted radiation from an ELED into a fiber. For example, in Figure 3.47 (a), a hemispherical lens attached to the fiber end is used for collimating the beam into the fiber. **A graded index (GRIN) rod lens** is a glass rod that has a parabolic refractive index profile across its cross-section with the maximum index on the rod axis. It is like a large diameter short length graded index "fiber" (typical diameters are 0.5–2 mm). A GRIN rod lens can be used to focus the light from an ELED into a fiber as illustrated in Figure 3.47 (b). This coupling is particularly useful for single mode fibers inasmuch as their core diameters are typically ~10 μm.

The output spectra from surface and edge-emitting LEDs using the same semiconductor material are not necessarily the same. The first reason is that the active layers have different doping levels. The second is the self-absorption of some of the photons guided along the active layer as in the ELED. Typically the linewidth of the output spectrum from an ELED is *less* than that from a SLED. For example, in a particular set of experiments on an InGaAsP ELED operating near 1300 nm, the emission linewidth was reported as 75 nm whereas the corresponding SLED at the same wavelength had a linewidth of 125 nm, which is significantly wider.

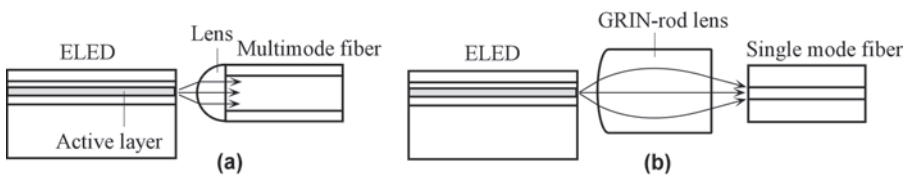


FIGURE 3.47 Light from an edge-emitting LED is coupled into a fiber typically by using a lens or a GRIN rod lens.

3.17 PHOSPHORS AND WHITE LEDS

Photoluminescence is the emission of light by a material, called a **phosphor**, that has been first excited by light of higher frequency; higher energy photons are first absorbed, and then lower energy photons are emitted. Typically the emission of light occurs from certain dopants, impurities, or even defects, called luminescent or **luminescence centers**, purposefully introduced into a **host matrix**, which may be a crystal or glass. The luminescent center is also called an **activator**. Many phosphors are based on activators doped into a host matrix; for example, Eu^{3+} (europium ion) in a Y_2O_3 (yttrium oxide) matrix is a widely used modern phosphor. When excited by UV radiation, it provides an efficient luminescence emission in the red (around 613 nm). It is used as the red-emitting phosphor in color TV tubes and in modern tricolor fluorescent lamps. Another important phosphor is Ce^{3+} in $Y_3Al_5O_{12}$ (YAG), written as $Y_3Al_5O_{12}$:Ce^{3+}, which is used in white LEDs. YAG:Ce^{3+} can absorb blue radiation and emit yellow light.

In very general terms, we can represent the energy of an activator in a host matrix by the highly simplified energy diagram in Figure 3.48 (a). We can take this figure to very roughly represent the energy of Ce^{3+} in YAG. Although it looks complicated, there are actually two manifolds of energies involved in blue absorption and yellow emission. There is a manifold of energies, labeled E_1, starting at E_1', which represents the ground energy of the activator. There is a higher manifold of energies, labeled E_2, starting at E_2'. A manifold such as E_1 has a number of energy levels that are closely spaced. The energy levels within the E_1- and E_2-manifolds represent the vibrational energies of the Ce^{3+}-ion in the host (YAG).[27] The slight horizontal shifts in the energy levels in a manifold in Figure 3.48 (a) indicate the relative position (or the tiny little displacement) of the Ce^{3+} ion in the host. Radiative transitions (absorption or emission of a photon) occur quickly compared with the time scale of vibrations of the Ce^{3+}-ion so that, during the radiative transition, the Ce^{3+} ion is essentially stationary. Thus, these radiative transitions are vertical lines in the activator energy diagram in Figure 3.48 (a); this rule is called the **Franck–Condon principle**.

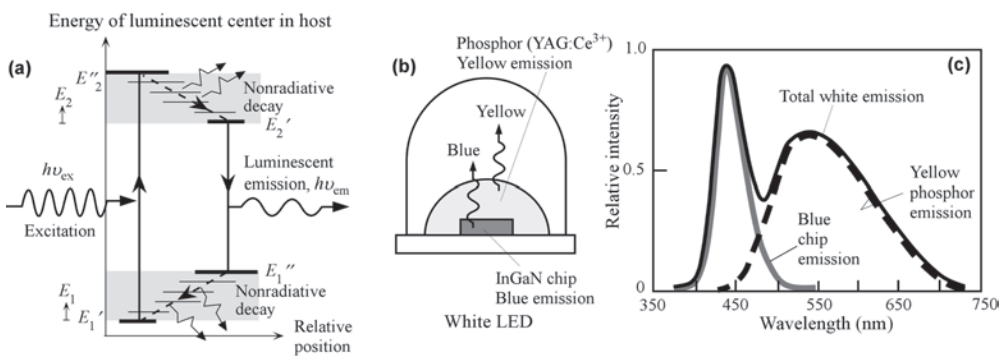


FIGURE 3.48 (a) A simplified energy diagram to explain the principle of photoluminescence. The activator is pumped from E_1' to E_2''. It decays nonradiately down to E_2'. The transition from E_2' down to E_1'' emits a photon with a lower energy than the excitation photon. (b) Schematic structure of a blue chip yellow phosphor white LED (c) The spectral distribution of light emitted by a white LED. Blue luminescence is emitted by GaInN chip and "yellow" phosphorescence is produced by phosphor. The combined spectrum looks "white."

[27] The energy diagram in Figure 3.48 (a) is still oversimplified. The vibrations of the Ce^{3+} ion is quantized and these energy levels are added to the usual electronic energy levels. However, there is also a dependence on the position of the Ce^{3+} ion, which has been simplified and represented as slightly shifted energy levels along the horizontal direction.

Upon excitation by an incident radiation of suitable energy $h\nu_{ex}$, the activator becomes excited to E_2'' in the E_2-manifold; a vertical transition. From this energy level, it decays, or *relaxes*, down relatively quickly (on a time scale of the order of picoseconds) to an energy level E_2' by emitting phonons or lattice vibrations. This type of decay is called **radiationless** or **non-radiative decay**. The Ce^{3+} ion becomes slightly displaced as shown in Figure 3.48 (a). From E_2', the activator decays down to E_1' in the E_1-manifold by emitting a photon by spontaneous emission, which is the emitted luminescent radiation. This is a vertical transition (Franck–Condon principle). The emitted photon energy is $h\nu_{em}$, which is less than the excitation photon energy $h\nu_{ex}$. The return from E_1' to the ground E_1 state involves phonon emissions. In some activators, the higher levels may form multilevel narrow energy "bands." In this example, the activator absorbed the incident radiation, and was directly excited, which is known as **activator excitation**. The Ce^{3+} ions in $Y_2Al_5O_{12}:Ce^{3+}$ can be excited directly by blue light, and would then emit in the yellow region as discussed below. It is apparent from Figure 3.48 (a) that the emitted radiation ($h\nu_{em}$) has a *longer* wavelength than the exciting radiation ($h\nu_{ex}$), that is, $h\nu_{em} < h\nu_{ex}$. The downshift in the light frequency from absorbed to emitted radiation is called the **Stokes shift**. It should be emphasized that the energy levels of the activator also depend on the host, because the internal electric fields within the host crystal act on the activator, and shift these levels up and down. The emission characteristics depend firstly on the activator, and secondly on the host.

Commercially available and popular white LEDs can provide bright light at a fraction of input electric power compared with conventional tungsten-based incandescent lights. In addition, the white LED has an extremely long lifetime. Most common white LEDs use a chip emitting at a short wavelength (blue or violet) and a *phosphor*, which absorbs some of the light from the blue diode and undergoes secondary luminescence emissions at a longer wavelength (yellow) as illustrated in Figure 3.48 (b). The mixture of blue light from the chip and yellow light from the phosphor results in an overall spectrum, shown in Figure 3.48 (c), that appears white to the eye. The quality and spectral characteristics of the combined emission in Figure 3.48 (c) vary with different white LED designs, for example, chip and phosphor combination.

Typically, a phosphor is composed of an inorganic host substance that contains optically active dopants; the dopants are the atoms that emit the lower energy photons, that is, the activators. Yttrium aluminum garnet (YAG) is a common host material. For white LED applications, it is usually doped with one of the rare-earth elements or a rare-earth compound. Cerium (Ce) is a common dopant element in YAG phosphors designed for white light–emitting diodes in



LUXEON Rebel ES white-emitting LED. *(Courtesy of Philips Lumileds.)*



Snap LED emitting in the amber for automotive signaling applications. The lamp includes the driver under the LED. *(Courtesy of Philips Lumileds.)*



The Audi A8 uses LEDs for nearly all its lighting, including headlights. *(Used with permission of Audi of America, Inc.)*

so-called **blue chip and yellow phosphor white LEDs**. White LEDs are now challenging the existing incandescent sources for general lighting.

In applications requiring a full spectrum of colors from a single point source, red, green, and blue (RGB)–emitting chips in a single package are used, which allows the generation of white light or any of 256 colors by utilizing circuits that drive the three diodes independently.

Additional Topics

3.18 LED ELECTRONICS

The electronics involved in driving LEDs depends on the application, which generally falls into three categories: (a) DC operation, (b) modulation of the LED output light intensity about some DC value in analog applications, and (c) pulsed operation in digital applications, including optical communications. In DC operation, the LED is driven by providing the necessary current for the required optical output. The *I–V* characteristics is such that large changes in the current correspond to small voltage changes across the LED as in Figure 3.43 (a), and for a given LED, V_F does not vary very much. The simplest circuit is shown in Figure 3.49 (a), where a DC voltage $V(>V_F)$ provides the current, which is determined by $I_F = (V - V_F)/R$. The problem is that keeping I_F constant depends on keeping V constant. Most LEDs have low reverse breakdown voltages and usually another diode, such as D, is used in parallel to shunt the LED when a reverse voltage is accidently applied to the LED.

A better circuit than that in Figure 3.49 (a) should provide the necessary drive current I_F to the LED, with some convenient way of controlling I_F. Bipolar junction transistors (BJTs) are commonly used in many applications as convenient current sources. Figure 3.49 (b) is shown as an example in which the LED is in the collector circuit of the BJT, which is driven by the output of an op amp A. The resistor R_F in the emitter of the BJT provides feedback, $I_F R_F$, to the negative input of A, which stabilizes the circuit. It is straightforward to show that $I_F \approx V/R_F$, that is, the circuit provides a linear control of I_F by the input voltage V. There are many ICs available that will perform the function of providing the necessary constant current under a wide range of input

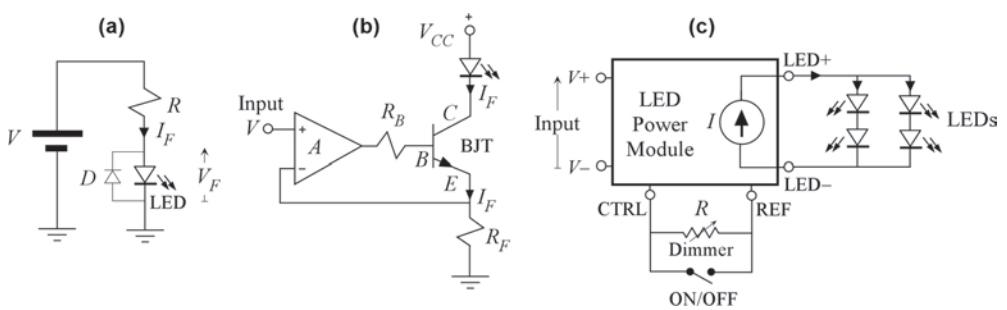


FIGURE 3.49 (a) The simplest circuit to drive an LED involves connecting it to a voltage supply (V) through a resistor R. (b). Bipolar junction transistors are well suited for supplying a constant current. Using an IC and negative feedback, the current is linearly controlled by V. (c) There are various commercial LED driver modules that can be easily configured to drive a number of LEDs in parallel and/or series. The example has a module driving four LEDs, a dimmer (R), and an on/off switch.

voltages, and temperature variations. Without a proper control of the current, the temperature variations can cause substantial changes in the current and hence the light output. Figure 3.49 (c) uses one of several LED driver modules available in the market, which can drive more than one LED in various parallel and series combinations, provided the maximum current rating is satisfied. The current can be externally and easily controlled, for example, by an external variable resistor, or a microprocessor.

The LED drive circuit shown in Figure 3.49 (b) can also be used in modulating the output of the LED as well since the current is proportional to the input voltage as in Figure 3.50 (a). One can, of course, design faster drive circuits, but the bottleneck will eventually be in the speed of response of the LED. An AC signal of frequency f on top of a DC voltage is used to drive the LED in Figure 3.50 (a). The frequency dependence of the output light power $P_o(f)$ with respect to its DC value $P_o(0)$ is shown in Figure 3.50 (b). There is a clear roll-off at high frequencies. The **cutoff frequency** f_c is defined as the frequency at which the normalized power drops by a factor of $2^{1/2}$, that is, at $f = f_c$, $P_o(f_c) = 0.707P_o(0)$. Any modulation of the LED current would result in modulating the minority carrier injection profile as shown in Figure 3.24 (a). The small signal equivalent circuit shown in Figure 3.24 (c) is usually sufficient to understand the frequency response of the LED but we have to be careful inasmuch as r_d and C_{diff} also depend on the frequency at high frequencies. First, consider very low frequencies, that is, almost DC. The capacitances are negligible. The diode action embedded in $I = I_o \exp(eV/k_BT)$ and hence the optical power output P_o, depend on the current in r_d. At high frequencies, the capacitances shunt away this current from r_d and decrease the optical output power. The characteristic time constant τ should be r_dC_{diff} (C_{diff} is greater than the depletion layer capacitance C_{dep}), that is, the net minority carrier recombination time. A more rigorous treatise arrives at the same conclusion and gives the output optical power $P_o(\omega)$ at frequency ω as

LED optical power output

$$P_o(\omega)/P_o(0) = 1/\left[1 + (\omega\tau)^2\right]^{1/2} \tag{3.18.1}$$

At the critical frequency $f_c = 1/(2\pi\tau)$ in Eq. (3.18.1), the relative output power is 0.707 as shown in Figure 3.50 (b). The **optical bandwidth** f_{op} is defined as the frequency at which $P_o(f_{\text{op}})/P_o(0) = 1/2$, and occurs when $f_{\text{op}} = \sqrt{3}f_c$, that is,

Optical bandwidth of an LED

$$f_{\text{op}} = \sqrt{3}/(2\pi\tau) \tag{3.18.2}$$

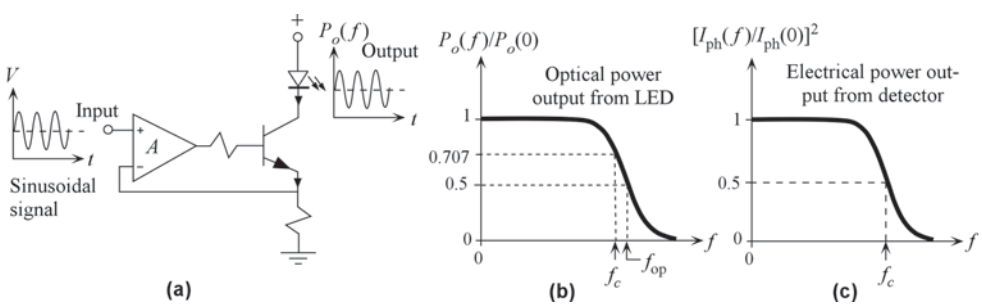


FIGURE 3.50 (a) Sinusoidal modulation of an LED. (b) The frequency response where f_c is the cutoff frequency at which $P_o(f)/P_o(0)$ is 0.707. (c) The electrical power output from the detector as a function of frequency. At f_c, $\left[I_{\text{ph}}(f)/I_{\text{ph}}(0)\right]^2$ is 0.5. However, it is 0.707 at a lower frequency than f_c.

where τ is the effective or net recombination time that is, $1/\tau = 1/\tau_r + \tau_{nr}$, where τ_r and τ_{nr} are the radiative and nonradiative recombination lifetimes. Recombination centers in the semiconductor reduce τ_{nr} and hence τ, which would widen the bandwidth. However, the IQE is also reduced, so that the optical output power is diminished as well, which is normally undesirable. If we feed the output from the LED to a photodetector, then the detector photocurrent I_{ph} will be proportional to P_o and the electrical power delivered by the detector will be proportional to I_{ph}^2 or P_o^2. At the frequency f_c, the detector's electrical output power will be proportional to $[0.707P_o(0)]^2$ or $0.5P_o(0)^2$. Thus, f_c represents a cutoff frequency (a bandwidth) at which the electrical power available from the detector has fallen to half its unmodulated value ($f = 0$ value), as illustrated in Figure 3.50 (c); f_c is called the **electrical bandwidth**.

The turning on and off of an LED in digital applications involves pumping the minority carriers into the active region and then removing them when the LED is turned off. In principle, the minority carrier lifetime τ should also play a key role in this pulsed operation. Figure 3.51 shows a simplified diagram for driving the LED digitally, for example, from a logic gate. The light output pulse should have an exponential rise and fall portions that delay the full turn-on and turn-off of the LED. The characteristic time constant for the rise and fall portions of the pulse should be the net recombination time τ. In practice, exponential rise and fall only serve as approximations to the switching behavior of the output light pulse. The net recombination time τ itself can depend on the current, and hence we should not expect a perfect exponential rise and fall.

Most LED data sheets quote the rise and fall times τ_R and τ_F for LEDs that have been pulsed. τ_R is the rise time from 10% to 90% of the final pulse height as shown in Figure 3.51. Similarly the fall time τ_F is the time it takes for the pulse to drop from its 90% to 10% value before the turn-off was triggered. If we approximate the rise and fall portions as exponentials as mentioned above, then, $\tau_R = \tau_F = 2.2\tau$. It is possible to shape the output pulse by intruding a pulse shaping capacitor C across the resistor R as shown in Figure 3.51. Upon triggering, the capacitor can provide the initial large current to pump (inject) the minority carriers in and hence turn on the device more quickly. Similarly, the capacitor would allow a large turn-off current to take out the minority carriers. Thus τ_R and τ_F can be shortened. The RC time constant must match the net recombination time in the LED for best reduction in the rise and fall times.

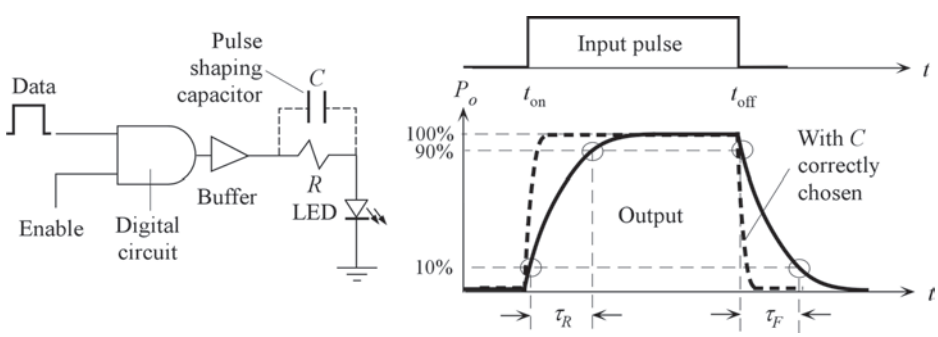


FIGURE 3.51 An LED in a digital circuit is turned on and off by a logic gate, assumed to have a buffered output as shown, to avoid being loaded. A BJT can be used after the logic gate to drive the LED as well (not shown). Definitions of rise and fall times are shown in the light output pulse.

Questions and Problems

3.1 Metals and work function The metal sodium (Na) has an atomic concentration of 2.54×10^{22} cm^{-3}. Each Na atom in isolation has one outer valence electron in an unfilled 3*s*-subshell. Once the solid is formed, the outer valence electrons from all Na atoms are shared in the crystal by all Na^+ ions, that is, these valence electrons find themselves in an unfilled energy band as in Figure 3.1 (c). Since each Na atom donates one electron to the energy band, the electron concentration is the same as the atomic concentration. Calculate the Fermi energy at 0 K. What is the speed of the electrons at E_F? What should be the speed at 300 K if the electrons could be treated classically in terms of the kinetic molecular theory as if they were free, similar to the atoms in a gas?

3.2 Photocathode and work function The photocathode of a photomultiplier tube has a multi-alkaline (Sb-Na-K-Cs) metal with a work function (Φ) of 1.55 eV. What is the longest wavelength that will cause photoemission? What is the kinetic energy of a photoemitted electron if the incident light wavelength is 450 nm (blue)? The quantum efficiency (QE) of a photocathode is defined by

$$\text{Quantum efficiency} = \frac{\text{Number of photoemitted electrons}}{\text{Number of incident photons}}$$

The QE is 100% if each incident photon ejects one electron. Suppose that blue light of wavelength 450 nm with an intensity of 1 μW cm^{-2} is incident on this photocathode with an area of 50 mm^2. If the emitted electrons are collected by applying a positive bias voltage to an anode, and the photocathode has a QE of 25%, what will be the photocurrent? (Normally the photoemitted electron is accelerated by a suitable applied field and impacts another electrode, a dynode, where it causes secondary electron emission, and so on, until the current is multiplied by orders of magnitude.)

3.3 Refractive index and bandgap Diamond, silicon, and germanium all have the same diamond unit cell. All three are covalently bonded solids. Their refractive indices (n) and energy bandgaps (E_g) are shown in Table 3.2. (a) Plot n vs. E_g and (b) Plot also n^4 vs. $1/E_g$. What is your conclusion? According to **Moss's rule**, roughly, $n^4 E_g \approx K$, a constant. What is the value of K?

TABLE 3.2 The refractive index n and the bandgap E_g of diamond, Ge, and Si, all of which have the same crystal structure

Material →	Diamond	Silicon	Germanium
Bandgap, E_g (eV)	5	1.1	0.66
n	2.4	3.46	4.0

3.4 Electrons in the CB of a nondegenerate semiconductor

(a) Consider the energy distribution of electrons $n_E(E)$ in the conduction band. Assuming that the density of state $g_{CB}(E) \propto (E - E_c)^{1/2}$ and using Boltzmann statistics $f(E) \approx \exp\left[-(E - E_F)/k_BT\right]$, show that the energy distribution of the electrons in the CB can be written as

$$n_x(x) = Cx^{1/2}\exp(-x)$$

where $x = (E - E_c)/k_BT$ is the electron energy in terms of k_BT measured from E_c and C is a constant at a given temperature (independent of E).

(b) Setting arbitrarily $C = 1$, plot $n_x(x)$ vs. x. Where is the maximum and what is the FWHM (full width at half maximum, that is, between half maximum points)? Is the use of $1.8k_BT$ for the half-maximum width correct?

(c) Show that the average electron energy in the CB is $(3/2)k_BT$, by using the definition of average,

$$x_{\text{average}} = \int_0^\infty x n_x dx \Big/ \int_0^\infty n_x dx$$

where the integration is from $x = 0$ (E_c) to say $x = 10$ (far away from E_c where $n_x \to 0$). You need to use a numerical integration.

(d) Show that the maximum in the energy distribution is at $x = (1/2)$ or at $E_{\max} = (1/2)k_BT$.

3.5 Intrinsic and doped GaAs The properties of GaAs are shown in Table 3.1. Calculate the intrinsic concentration and the intrinsic resistivity at room temperature (take as 300 K). Where is the Fermi level? Assuming the N_c and N_v scale as $T^{3/2}$, what would be the intrinsic concentration at 100°C? If this GaAs crystal is doped with 10^{17} donors cm^{-3} (such as Te), where is the new Fermi level and what is the resistivity of the sample? The drift motilities in GaAs are shown in Table 3.3.

TABLE 3.3 Ionized dopant impurities scatter carriers and reduce the drift mobility. The dependence of μ_e for electrons and μ_h for holes on the total ionized dopant concentration

Dopant concentration (cm^{-3})	0	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
GaAs, μ_e ($cm^2\ V^{-1}\ s^{-1}$)	8500	–	8000	7000	5000	2400
GaAs, μ_h ($cm^2\ V^{-1}\ s^{-1}$)	400	–	380	310	250	160
Si, μ_e ($cm^2\ V^{-1}\ s^{-1}$)	1450	1420	1370	1200	730	280
Si, μ_h ($cm^2\ V^{-1}\ s^{-1}$)	490	485	478	444	328	157

3.6 Electrons in GaAs Given that the electron effective mass m_e^* for the GaAs is $0.067m_e$, calculate the thermal velocity of the electrons in the CB of a nondegenerately doped GaAs at room temperature (300 K). If μ_e is the drift mobility of the electrons and τ_e the mean free time between electron scattering events (between electrons and lattice vibrations) and if $\mu_e = e\tau_e/m_e^*$, calculate τ_e, given $\mu_e = 8500\,cm^2\ V^{-1}\ s^{-1}$. Calculate the **drift velocity** $v_d = \mu_e E$ of the CB electrons in an applied field E of $10^5\ V\ m^{-1}$. What is your conclusion?

3.7 Extrinsic *n*-GaAs An *n*-type GaAs crystal is doped with 10^{16} donors cm^{-3} (such as Te), what are the electron and hole concentrations, and the conductivity? (See Table 3.3.)

3.8 Extrinsic *n*-Si A Si crystal has been doped *n*-type with $1 \times 10^{17}\,cm^{-3}$ phosphorus (P) donors. The electron drift mobility μ_e depends on the total concentration of ionized dopants N_{dopant}, as in Table 3.3, inasmuch as these ionized dopants scatter the electrons and thereby decrease their drift mobility. What is the conductivity of the sample? Where is the Fermi level with respect to the intrinsic crystal?

3.9 Compensation doping in *n*-type Si An *n*-type Si sample has been doped with 10^{16} phosphorus (P) atoms cm^{-3}. (a) What are the electron and hole concentrations? (b) Calculate the room temperature conductivity of the sample. (c) Where is the Fermi level with respect to E_{Fi}? (d) If we now dope the crystal with 10^{17} boron acceptors, what will be the electron and hole concentrations? (e) Where is the Fermi level with respect to E_{Fi}?

3.10 Free carrier absorption in semiconductors The attenuation of light due to the optical field drifting the free carriers is called **free carrier absorption**. As the free electrons in a semiconductor crystal are accelerated by the optical field, they eventually become scattered by lattice vibrations or impurities, and pass the energy absorbed from the radiation to lattice vibrations. In such cases, ε_r'' and the AC conductivity σ at the same frequency are related by

$$\varepsilon_r'' = \sigma/\varepsilon_o\omega \tag{P3.1}$$

We consider a semiconductor in which the free carriers are electrons (an *n*-type semiconductor). The AC conductivity σ in general is given by

$$\sigma = \sigma_o/(1 + j\omega\tau_e) \tag{P3.2}$$

where σ_o is the DC conductivity, ω is the angular frequency of light, and τ_e is the scattering time of the conduction electrons. σ decreases with frequency in Eq. (P3.2). Consider an *n*-type semiconductor. The free carriers are the electrons in the CB. If the drift mobility of the electrons is μ_e, then $\mu_e = e\tau_e/m_e^*$, where m_e^* is the effective mass of the electrons in the CB of the semiconductor.[28] The DC conductivity $\sigma_o = en\mu_e$, where n is the concentration of CB electrons. Show that the absorption coefficient due to free carrier absorption (due to the conductivity) when $\omega > 1/\tau$ is given by

$$\alpha = \left(\frac{\sigma_o}{nc\varepsilon_o\tau^2}\right)\left(\frac{1}{\omega^2}\right) = \left(\frac{e^3 n}{4\pi^2 nc^3\varepsilon_o m_e^{*2}\mu_e}\right)\lambda^2 \tag{P3.3}$$

[28] As we know, conduction in an *n*-type semiconductor occurs by the drift of free electrons inside the semiconductor crystal. The dopants donate electrons to the crystal, which are free within the crystal.

where n is the refractive index. What would you expect if you plotted α vs. λ^2? Consider a Si crystal doped with 10^{15} cm^{-3} donors. Estimate the free carrier absorption (in m^{-1} and dB m^{-1}) at 1.55 and at 5 μm. What is your conclusion? [Although Eq. (P3.3) is a highly simplified approximation to describing free carrier absorption, it nonetheless provides a rough estimate of its magnitude.]

3.11 GaAs *pn* junction Consider a GaAs *pn* junction that has the following properties: $N_a = 10^{16}$ cm^{-3} (p-side); $N_d = 10^{18}$ cm^{-3} (n-side); $B = 2.0 \times 10^{-16}$ m^3 s^{-1}; cross-sectional area $A = 1.5$ mm $\times$ 1.5 mm. Assume a long diode. What is the diode current due to diffusion in the neutral regions and recombination in the SCL at 300 K when the forward voltage across the diode is 0.8 V and then 1.1 V? (Use the drift motilities in Table 3.3 for calculating the diffusion coefficients through the Einstein relation.)

3.12 InP *pn* junction Consider an InP *pn* junction that has the following properties: $N_a = 10^{15}$ cm^{-3} (p-side); $N_d = 10^{17}$ cm^{-3} (n-side); using $B \approx 4 \times 10^{-16}$ m^3 s^{-1}; cross-sectional area $A = 1$ mm $\times$ 1 mm. Assume a long diode. What is the diode current due to diffusion in the neutral regions and recombination in the SCL at 300 K when the forward voltage across the diode is 0.70 and 0.9 V? The electron mobility in the p-side is about ~6000 cm^2 s^{-1} and the hole mobility on the n-side is roughly ~100 cm^2 s^{-1}. (See also Table 3.1 for n_i and ε_r.) Comment on the ideality factor of this InP *pn* junction.

3.13 Si *pn* junction Consider a long *pn* junction diode with an acceptor doping N_a of 10^{18} cm^{-3} on the p-side and donor concentration of N_d on the n-side. The diode is forward biased and has a voltage of 0.6 V across it. The diode cross-sectional area is 1 mm^2. The minority carrier recombination time, τ, depends on the total dopant concentration, N_{dopant}(cm^{-3}), through the following approximate empirical relation

$$\tau \approx (5 \times 10^{-7})/(1 + 2 \times 10^{-17} N_{\text{dopant}})$$

where τ is in seconds.

(a) Suppose that $N_d = 10^{15}$ cm^{-3}. Then the depletion layer extends essentially into the n-side and we have to consider minority carrier recombination time, τ_h, in this region. Calculate the diffusion and recombination contributions to the total diode current given $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3}. Use Table 3.3 for μ_e and μ_h. What is your conclusion?

(b) Suppose that $N_d = N_a$. Then W extends equally to both sides and, further, $\tau_e = \tau_h$. Calculate the diffusion and recombination contributions to the diode current given $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{18}$ cm^{-3}. Use Table 3.3 for μ_e and μ_h. What is your conclusion?

3.14 Injected minority carrier charge Consider a *pn* junction with heavier doping on the p-side. The injected minority carriers (holes) represent an *injected excess minority carrier charge* Q_h in the neutral region as shown in Figure 3.24 (a). (There is also excess majority carrier charge so the region is neutral.) Show that

$$Q = I\tau_h \text{ for a long diode and } Q = I\tau_t \text{ for a short diode}$$

in which τ_h is the hole lifetime and τ_t is the diffusion time, or the transit time of holes across the width of the neutral n-region, that is, $\tau_t = l_n^2/2D_h$. What is your conclusion?

3.15 High injection condition The Shockley equation for a *pn* junction under forward bias, as shown in Figure 3.16 (a), was derived by assuming low (weak) injection conditions, that is $p_n(0) \approx \Delta p_n(0) \ll n_{no}$ or N_d on the n-side. Show that when the injection is no longer weak, that is when $p_n(0) \approx n_{no} = N_d$, the applied voltage V reaches V_{SI} (strong injection) given by

$$V_{\text{SI}} = V_o - V_{\text{th}} \ln(N_a/N_d)$$

where V_{th} is the thermal voltage (k_BT/e). Calculate V_o and V_{SI} for a Si *pn* junction that has $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{16}$ cm^{-3}. Can you use the Shockley equation when $V > V_{\text{SI}}$? What happens when $N_a = N_d$? What is your conclusion?

3.16 Heterostructure Consider a Type I heterostructure as shown in Figure 3.27.

(a) If $E_{g1} < E_{g2}$ and if χ_1 and χ_2 are the electron affinities of each semicondcutor, show that

$$\Delta E_c = \chi_1 - \chi_2 \quad \text{and} \quad \Delta E_v = E_{g2} - E_{g1} - \Delta E_c$$

(b) Using the data in Table 3.1, draw the energy band diagram of an *nP* junction between an n-type Ge and P-GaAs. Under forward bias, is it easier to inject electrons or holes?

(c) Draw the energy band diagram for a *pN* junction between p-type Ge and N-GaAs. Under forward bias, is it easier to inject electrons or holes?

3.17 Heterojunction *I–V* characteristics We use some of the data reported by Womac and Rediker (*J. Appl. Phys.*, *43*, 4130, 1972) for AlGaAs/GaAs *pn* heterojunction at 298 K (25°C). Sample A is an N^+p and sample B is a P^+n junction. The *I–V* data of interest are listed in Table 3.4. By a suitable plot find the ideality factor for each. What is your conclusion?

TABLE 3.4 *I–V* **data on two heterojunctions. First set is N^+p and the second set is a P^+n junction**

A: Heterojunction N^+p														
V	0.206 V	0.244	0.290	0.322	0.362	0.412	0.453	0.485	0.537	0.576	0.612	0.662	0.708	
I	1.03 nA	2.07	5.20	10.3	20.7	52.8	105	192	515	1.02 μA	2.03	4.89	10.1	
B: Heterojunction P^+n														
V	0.310 V	0.364	0.402	0.433	0.485	0.521	0.561	0.608	0.682	0.726	0.764	0.807	0.859	0.885
I	2.01 nA	4.91	9.79	19.0	49.5	96.1	194	466	1.96 μ A	5.02	9.75	19.5	50.6	99.5

3.18 AlGaAs LED emitter An AlGaAs LED emitter for in a local optical fiber network has the output spectrum shown in Figure 3.32 (b). It is designed for peak emission at about 822 nm at 25°C. (a) Why does the peak emission wavelength increase with temperature? (b) What is the bandgap of AlGaAs in this LED? (c) The bandgap, E_g, of the ternary $Al_xGa_{1-x}As$ alloys follows the empirical expression, $E_g(eV) = 1.424 + 1.266x + 0.266x^2$. What is the composition of the $Al_xGa_{1-x}As$ in this LED?

3.19 III–V compound semiconductors in optoelectronics Figure 3.52 represents the bandgap E_g and the lattice parameter a in a quaternary III–V alloy system. A line joining two points represents the changes in E_g and a with composition in a ternary alloy composed of the compounds at the ends of that line. For example, starting at GaAs point, $E_g = 1.42$ eV and $a = 0.565$ nm, E_g decreases and a increases as GaAs is alloyed with InAs, as we move

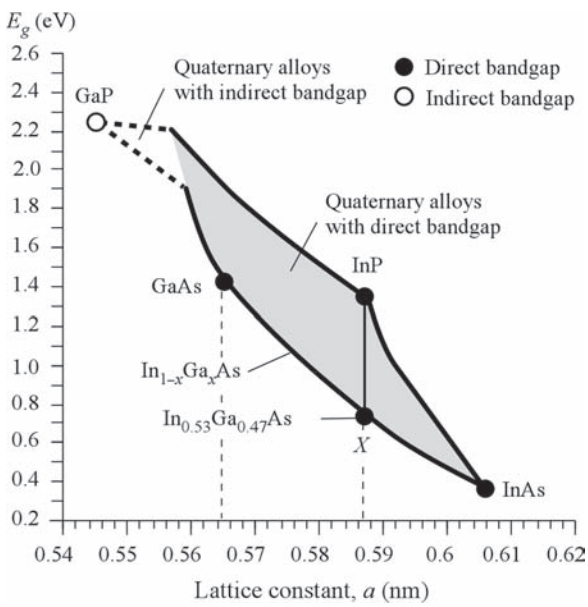
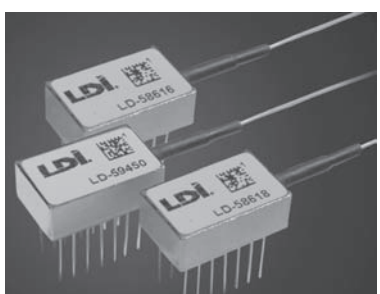


FIGURE 3.52 Bandgap energy E_g and lattice constant a for various III–V alloys of GaP, GaAs, InP, and InAs. A line represents a ternary alloy formed with compounds from the end points of the line. Solid lines are for direct bandgap alloys whereas dashed lines for indirect bandgap alloys. Regions between lines represent quaternary alloys. The line from X to InP represents quaternary alloys $In_{1-x}Ga_xAs_{1-y}P_y$ made from $In_{0.53}Ga_{0.47}As$ and InP, which are lattice-matched to InP.



InGaAsP 1300 nm LED emitters, each pigtailed to an optical fiber for use in optical communication modems and lower speed data/analog transmission systems. *(Courtesy of OSI Laser Diode, Inc.)*

along the line joining GaAs to InAs. Eventually at InAs, $E_g = 0.35\,\text{eV}$ and $a = 0.606\,\text{nm}$. Point X in Figure 3.52 is composed of InAs and GaAs and it is the ternary alloy $In_{1-x}Ga_xAs$. At X, $In_{0.53}Ga_{0.47}As$ (often called "in-gas" in telecom) has $E_g = 0.73\,\text{eV}$ and $a = 0.587\,\text{nm}$, which is the same a as that for InP. $In_{1-x}Ga_xAs$ at X is therefore lattice-matched to InP and can hence be grown on an InP substrate without creating defects at the interface.

Further, $In_{1-x}Ga_x$ As at X can be alloyed with InP to obtain a quaternary alloy[29] $In_{1-x}Ga_xAs_{1-y}P_y$ whose properties lie on the line joining X and InP and therefore all have the same lattice parameter as InP but different bandgap. Layers of $In_{1-x}Ga_xAs_{1-y}P_y$ with composition between X and InP can be grown epitaxially on an InP substrate by various techniques such as liquid phase epitaxy (LPE) or molecular beam epitaxy (MBE). The grey shaded area between the solid lines represents the possible values of E_g and a for the quaternary III–V alloy system in which the bandgap is direct and hence suitable for direct recombination. The compositions of the quaternary alloy lattice matched to InP follow the line from X to InP.

(a) Given that the $In_{1-x}Ga_x$ As at X is $In_{0.53}Ga_{0.47}As$ show that quaternary alloys $In_{1-x}Ga_xAs_{1-y}P_y$ are lattice matched to InP when $y = 1 - 2.13x$.

(b) The bandgap energy E_g, in eV for $In_{1-x}Ga_xAs_{1-y}P_y$ lattice-matched to InP is given by the empirical relation, $E_g(\text{eV}) = 0.75 + 0.46y + 0.14y^2$. Find the composition of the quaternary alloy suitable for an LED emitter operating at 1.30 μm.

3.20 Varshni equation and the change in the bandgap with temperature The Varshni equation describes the change in the energy bandgap E_g of a semiconductor with temperature T as given by Eq. (3.11.2) that is

$$E_g = E_{go} - AT^2/(B + T)$$

where E_{go} is E_g at 0 K, and A and B are constants. Show that

$$\frac{dE_g}{dT} = -\frac{AT(T + 2B)}{(B + T)^2} = -\frac{(E_{go} - E_g)}{T}\left(\frac{T + 2B}{T + B}\right)$$

For GaAs, $E_{go} = 1.519\,\text{eV}$, $A = 5.41 \times 10^{-4}\,\text{eV K}^{-1}$, $B = 204$ K. What is dE_g/dT for GaAs? Find the shift in the emitted wavelength from a GaAs LED per 1°C change at room temperature (300 K). Find the emission wavelength at 27°C and −30°C.

3.21 Dependence on the emission peak and linewidth on temperature Using the Varshni equation find the peak emission wavelength and the linewidth of the emission spectrum from an $In_{0.47}Ga_{0.53}As$ LED when it is cooled from 25°C to −25°C. You can use Eqs. (3.3.1) and (3.3.2). The Varshni constants for $In_{0.47}Ga_{0.53}As$ are $E_{go} = 0.814\,\text{eV}$, $A = 4.906 \times 10^{-4}\,\text{eV K}^{-1}$, $B = 301$ K.

3.22 LED Output Spectrum Given that the width of the relative light intensity vs. photon energy spectrum of an LED is typically around ~3 k_BT, calculate the spectral width in wavelength (nm) of LED emitters operating at 850 nm, 1310 nm, and 1550 nm?

3.23 Linewidth of LEDs Experiments carried out on various direct bandgap semiconductor LEDs give the output spectral linewidth (between half intensity points as listed in Table 3.5). From Figure 3.31 we know that a spread in the wavelength is related to a spread in the photon energy, $\Delta\lambda \approx (hc/E_{\text{ph}}^2)\Delta E_{\text{ph}}$ where $E_{\text{ph}} = h\upsilon$ is the photon energy. Suppose that we write $E_{\text{ph}} = hc/\lambda$ and $\Delta E_{\text{ph}} = \Delta(h\upsilon) \approx mk_BT$ where m is a numerical constant. Therefore,

$$\Delta\lambda \approx (mk_BT/hc)\lambda^2 \qquad \textbf{(P3.4)}$$

By appropriately plotting the data in Table 3.5, and assuming $T = 300\,\text{K}$, find m.

TABLE 3.5 Linewidth $\Delta\lambda_{1/2}$ between half points in the output spectrum (spectral intensity vs. wavelength) of eight LEDs using direct bandgap semiconductors

Material (Direct E_g)	AlGaAs	AlGaAs	AlGaAs	GaAs	GaAs	InGaAsP	InGaAsP	InGaAsP
Peak wavelength of emission (λ) nm	650	810	820	890	950	1150	1270	1500
$\Delta\lambda_{1/2}$ nm	22	36	40	50	55	90	110	150

[29] Some books have other formats for the chemical composition for example, $Ga_xIn_{1-x}As_yP_{1-y}$. The present notation $In_{1-x}Ga_xAs_{1-y}P_y$ was chosen to reflect the common vernacular for InGaAs (pronounced "in-gas").

Table 3.6 gives the linewidth $\Delta\lambda_{1/2}$ for various visible LEDs based on GaAsP. Radiative recombination is obtained by appropriately doping the material. Using $m = 3.0$ in Eq. (P3.4), $T = 300$ K, calculate the expected spectral width for each and compare with the experimental value. What is your conclusion?

TABLE 3.6 Linewidth $\Delta\lambda_{1/2}$ between half points in the output spectrum (intensity vs. wavelength) of various visible LEDs using GaAsP

Peak wavelength of emission (λ) nm	565	583	600	635
$\Delta\lambda_{1/2}$ nm	28	36	40	40
Color	Green	Yellow	Orange	Red
Material	GaP(N)	GaAsP(N)	GaAs (N)	GaAsP

3.24 Quantum wells We will consider a quantum well in a semiconductor heterostructure, and assume, for simplicity, an infinite potential energy quantum well with dimensions d along x, and D_y and D_z along y and z directions. The energy of an electron with respect to the bottom of the well is then given by

$$\Delta E = \frac{h^2 n^2}{8m_e^* d^2} + \frac{h^2 n_y^2}{8m_e^* D_y^2} + \frac{h^2 n_z^2}{8m_e^* D_z^2} \qquad \textbf{(P3.5)}$$

Energy in a quantum well

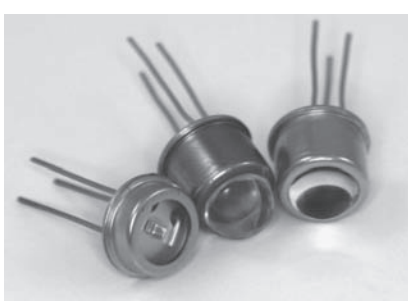
where n, n_y, and n_z are quantum numbers having the values 1, 2, 3, … Consider a well that has $d = 10$ nm, $D_x = D_y = 2\,\mu$m, and $m_e^* = 0.067m_e$. (a) Calculate the minimum energy. Which is the most significant contributing term in Eq. (P3.5) to the lowest energy? (b) What would n_y need to be to get the same energy as the first term? (c) What is the separation δE between the energy levels for motion in the y and z plane? What is your conclusion?

3.25 Energy levels in a quantum well Consider a GaAs QW sandwiched between two $Al_{0.40}Ga_{0.60}As$ layers. The barrier height ΔE_c is approximately 0.30 eV, the electron effective mass in the well is $0.067m_e$, and the width of the QW (d) is 8 nm. Calculate the energy levels ΔE_1 and ΔE_2 from the bottom of E_c, assuming an infinite PE well. Compare these with the calculations for a finite PE well that give $\Delta E_1 = 0.050$ eV and $\Delta E_2 = 0.197$ eV. What is your conclusion?

3.26 QW LED Linewidth Consider the QW-based InGaN (green to blue) and AlGaN (UV) LEDs whose peak emission wavelengths (λ_o) and linewidths (between half intensity points, $\Delta\lambda$) are listed in Table 3.7. For each LED, apply the linewidth equation $\Delta\lambda \approx (mk_BT/hc)\lambda_o^2$ at 300 K, and find m. Do you think this formula can be used for QW LEDs?

TABLE 3.7 Avago ASMT 1W series of InGaN green, cyan, blue, and royal blue LEDs and SETi AlGaN UV LED

LED	Green	Cyan	Blue	Royal blue	UV
λ_o	519	497	454	450	270
$\Delta\lambda$	31.25	29	22.5	20	12



UV emitting LEDs *(Courtesy of Sensor Electronic)*

3.27 External conversion efficiency The total light output power, called the radiant flux, from a number of commercial LEDs is listed in Table 3.8. The table also lists the emitted peak wavelength and the measured linewidth for each LED. For each device calculate the external quantum efficiency (%), EQE, and m in $\Delta\lambda \approx (mk_BT/hc)\lambda^2$. What is your conclusion?

TABLE 3.8 Characteristics of selected commercial LEDs covering the IR, blue, and UV

	IR, Hamamatsu L10823 (InGaAs)	IR, Hamamatsu L10822 (InGaAs)	IR, Osram SFH406 (GaAs)	IR, Vishay TSAL6100 (GaAlAs)	IR, Hamamatsu L1909 (AlGaAs)	Super Blue, Cree (InGaN)	UV, UVTOP 315, SETi (InGaN)
V_F (V)	0.9	1.1	1.25	1.35	1.4	3.8	5.5
I_F (mA)	50	50	40	100	50	30	20
P_o (mW)	2	3.1	7	35	10	1.15	0.6
λ_o (nm)	1650	1300	950	940	870	466	315
$\Delta\lambda$ (nm)	120	90	55	50	80	65	11

3.28 Light extraction efficiency of LEDs Extraction efficiency or ratio depends on the LED structure and the means that have been used to manipulate light to escape from the chip.

(a) Figure 3.53 (a) shows a bare LED chip in which the light from the active region is incident on the semiconductor–air interface. The refractive indices of the semiconductor and ambient are n_s and n_a respectively. Assume a GaN crystal for the semiconductor with a refractive index of 2.5. For normal incidence, what is the transmittance T from the semiconductor to the air? What is the critical angle θ_c at this interface? The fraction of output light from the active region within the emission cone defined by the critical angle θ_c has been evaluated to be $(1/2)[1 - \cos\theta_c]$. Taking into account T as well, estimate the extraction efficiency (EE) if the ambient is air, $n_a = 1$.

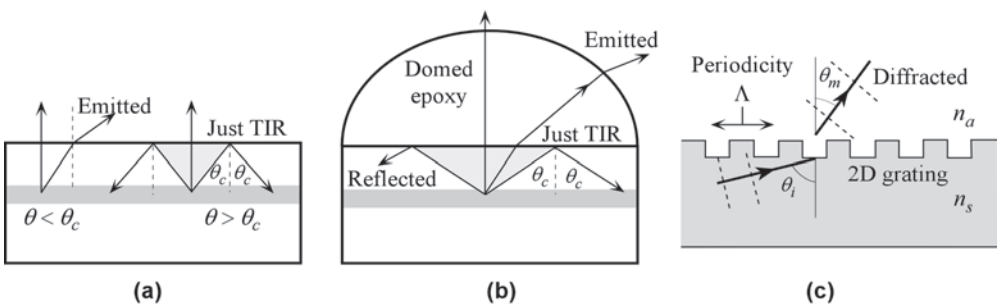


FIGURE 3.53 (a) Total internal reflection results in poor light extraction (b) Improvement in the light extraction by the use of a domed epoxy. (c) A two-dimensional diffraction grating (a photonic crystal) based surface enhances the extraction of light through Bragg diffraction.

(b) Figure 3.53 (b) shows an epoxy dome placed on top of the semiconductor. Suppose that the refractive index of the epoxy is 1.8. Calculate the new critical angle at the semiconductor–epoxy interface and estimate the new EE. What is the critical angle for light rays at the epoxy–air interface? How critical is it to have an exactly hemispherical surface?

(c) Figure 3.53 (c) shows an LED chip whose surface has been patterned to form a two-dimensional (2D) *photonic crystal*, which is a periodic modulation of the refractive index. With the right periodicity Λ, the photonic crystal will leak out the light trapped in the crystal. We will take a highly simplified view and treat the periodic variation as a grating. If θ_i is the angle of incidence, there is a diffracted beam at an angle θ_m, which satisfies the Bragg diffraction condition that involves two different media that is,

$$\Lambda(n_s \sin\theta_i - n_a \sin\theta_m) = m\lambda$$